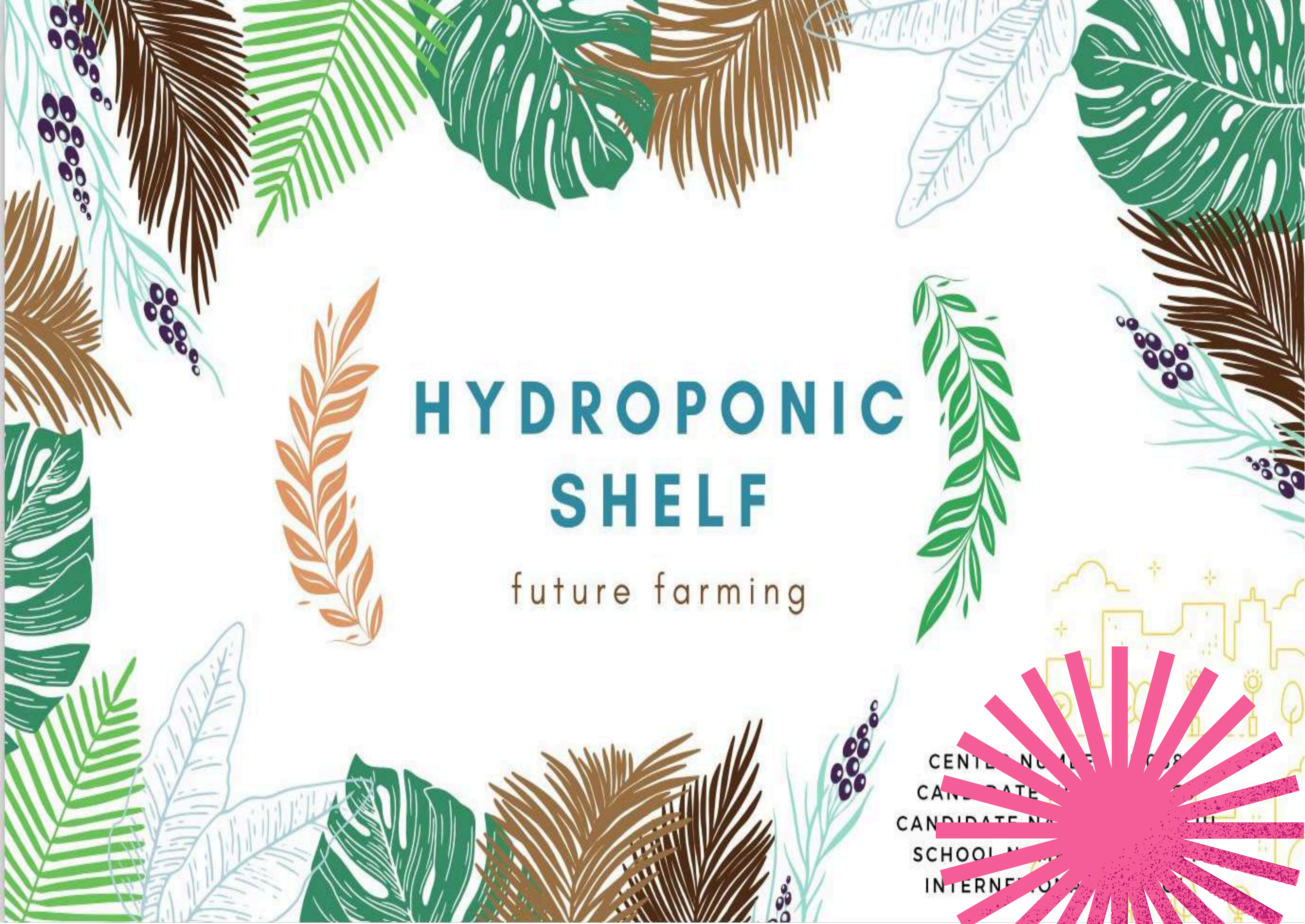


A decorative border of various tropical leaves and plants, including monstera, palm, and ferns, in shades of green, brown, and orange, framing the central text.

HYDROPONIC SHELF

future farming

A large, bright pink starburst or sunburst graphic with many sharp points, partially obscuring the text in the bottom right corner.

CENTER NUMBER FOR
CANDIDATE NUMBER
CANDIDATE NAME
SCHOOL NAME
INTERNET FOR

muppy Hanp



Design Brief



After comparing, I'd like to try to combine the advantage of hydroponics and aquaponics together since they are similar in structures and systems. Also, they will not make people's hands dirty, which can save water and time.

MATERIALS:

I believe a mix of plastic, wood and metal can present a functional, sturdy and nice-looking shelf.

EGRONOMICS:

The shelf should fit people's height. It might need a bigger place for plants to grow and for humans to pick.



AESTHETICS:

It should perform a minimalist and organic style with simple assembling procedures. It should be light in weight and fit in a small place, such as an apartment house. The highlight of this shelf is its multi-functional property, which can solve other trials in life. The colour scheme can be various. My investigation will base on these properties.



PROBLEMS:

Following the **exponentially increased population**, the **cities are expanding** every year and **people from poor places can't even fill their stomach** while **others are still wasting food**. Concrete buildings, highways, bridges and countless other factors affect both animal habitats and Earth climates. **The farming lands are disappearing(followed by more deforestation)**, especially in megacities around the world, making **people worried about pollution** and **the safety of food**(agriculture chemicals). I am from Hohhot in China. Whenever I read pollution news about Beijing, Shanghai, I always want to find a better way to slow down everything. Apart from that, **constant global warming** is another unavoidable problem due to the development of technology and human habitation.

Thus, we must do something. Having green plants to play their part, we can control most of the problems that we are facing nowadays.

DESIGN NEEDS:

Since the number of farmlands is declining, we have to think about **other ways of planting food**. I get an idea that what if we rely on fruits and vegetables that are planted in cities, but not in the actual farmland. Cities are increasing in size, so why don't we combine farmlands and cities? As you might know, some countries around the world already have the ideas of 'eco-friendly constructions' such as 'Gardens By The Bay in Singapore'. However, it is hard for us to give enough space only for the plants to grow. Some poor areas even can't afford it. The limitation of space is also a problem for people who love gardening but cannot have their garden in their flat.

When I visited Central, a popular department store in Thailand, I noticed its interior design is super 'fresh'. They use exotic flora as well as local plants to decorate the store which also can purify the air. Suddenly, I realized another way of placing farmlands--we can separate them to every single civils' houses. Bearing this thinking in mind, I investigate many popular and efficient way of planting plants indoor: **Container Gardening, Greenhouse Gardening, Hydroponics, and Aquaponics**. A space-saving shelf might be a better choice as an alternative to the farmland.

TYPES	DIFFICULTY	NEEDS OF SPACE	OTHER REQUIREMENTS	PRODUCTION	NEEDS OF WATER	other pros and cons
container	low	less space	soil, pots, fertilizer, pesticides	low	medium	dirty hands and house, semilly fertilizer
greenhouse	high	large space	soil, fertilizer, pesticides	high	high	temperature control needed
hydroponics	medium	less space	reservior, nutrient solution and specific equipments	high	medium	less water but high production
aquaponics	highest	less space	reservior, fish, fish food and specific equipments	medium	high	no need of nutrient--fishes' faeces benefits plants

FUNCTIONS & PROPERTIES:

Primary functions of the shelf:

1. Giving people a consistent amount of **fresh food** in a limited area.
2. Solving the problem of decreasing farmlands and forests.
3. Purifying the surrounding air.

Secondary functions of the shelf:

1. Putting other utensils
2. Creating a new way of studying plants at home
3. Close to nature--dragging the human and nature closer.

TARGET MARKET:

This design is typically for people:

- lives in, for examples, metropolitan area, does not have their garden, but they are desire to have one.
- has a small room for planting.
- wants to have fresh air at home.
- pursues the quality of life, wanting to improve their daily diet(e.g. organic food).
- loves plants, finding ways to approach nature
- lives in a food shortage area.
- wishes to improve their inner environment.



Research



systems

material

target market

6 MAIN SYSTEMS:

THEIR FUNCTION,

APPEARANCE,

COMPONENTS(COS

T), ADVANTAGES

AND

DISADVANTAGES.

WOODS: SOFTWOOD(PINE, FIR,

SPRUCE), HARDWOOD(MERANTI,

CHENGAL, RUBBER WOOD) AND

MANUFACTURED WOOD(PLYWOOD,

MDF, HARDBOARD)

METAL: FERROUS METALS AND

NON-FERROUS METALS

PLASTIC: THERMOPLASTIC AND

THERMOSETTING PLASTICS

MASS MARKET

HOBBIES: LOVE GARDENING,

GROWING THEIR OWN FOOD AT HOME.

DIFFERENT AGE GROUP:

TEENAGERS, ADULTS AND ELDERLY

GROUPS

Health and
sustainable
materials

SEARCHING PROBLEMS THAT
THE MATERIAL COULD CAUSE

METHOD OF MEASURING

using rulers, pencils, try squares,
marking gauges, etc.

Manufacture

METHOD OF SHAPING

using different types of saws, portable
power tools, sanding/polishing tools and
CNC machines(laser cutters& router)

METHOD OF FORMING

using 3D printers, injection moulding,
and line-bending

CAD+CAM

3D MODELING AND 2D DRAWING
CAN BE CREATE ONLINE!!!

METHOD OF ASSEMBLING

using various joins in wood(e.g. dowel, finger,
mitre, butt, mortise & tenon joints), understanding
soldering in metal, adhesive and other
manufacturing components(e.g.screws, nails,
hinges)

METHOD OF FINISHING

understand different finishing
such as oils, paints, stains,
choosing one suitable way(s) at
last.

HYDROPONIC SHELF

google
&
consultation

survey+comp
-arison

THE FINAL SHAPE CAN BE CHANGED
SINCE IT IS A DISMOUNTING SHELF

the **LOWEST** price
such as

around 200 bahts
can be used as a



COST

The cost of the project is different
since it depends on the materials
and people's demanding. Nutrients,
water and seeds are generally
cheap, so everyone can have one of
this shelf at home.

USAGES

This shelf can be placed
either indoor(e.g.in the
living room) or
outdoor(e.g.in the
balcony), and its purpose
is solving global
problems and providing
green food. It uses less
resource than traditional
potted plants, which also
give a good-looking at
home.



the **Highest** price
such as

around 5000 bahts



Inspiration

DESIGN



aesthetic

colour schemes: Natural,

Simplicity, Colorful, Original

colour of materials

Style : Simple, Geometrical,

modern, Natural form

Property: Practical, Decorative,

Multi-functional, Fruitful

size

Base on anthropometric and

ergonomics:

• flood-based

• desktop

• hanging type

[illegible]

WEEKS	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Design brief																									
Mind map																									
Gantt chart																									
Survey and Consultation																									
Mood board																									
Products analysis																									
Environmental research																									
Systems research																									
Joints research																									
Material research																									
Conclusion																									
Design specification																									
Initial ideas																									
Final design																									
Interview																									
Modelling																									
Testing of material																									
Construction decision																									
Product manufacture																									
Evaluation																									

//my conclusion

1.How old are you?

A.under 10 **B.under 20**
C.under 30 **D.more than 30**

//more adults are interested in it, so
the colour and shape should be more
mature.

2.Do you love plants or
gardening?

A.Yes **B.No** **C. Sometimes**

//people who love plants are more
interesting on the plant shelf.(target
market)

3.Where do you currently live?

A. city **B.A town** **C.A megacity**

//Urban civils is more interest in the
plant shelf(target market)

4. What is your gender?

A. Female **B. Male**

C. does not prefer to tell

//the female tend to love plants over
the male(target market)

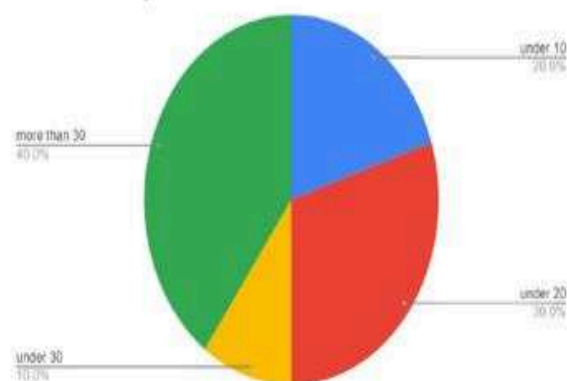
5.What types of colour scheme
do you like?

A.Natural colour **B.Simplicity**

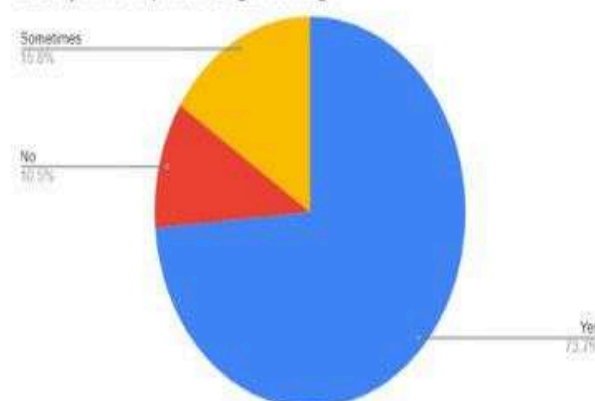
C.Original colour of materials **D.Colorful**

//natural colour is popular than
others(green,blue,yellow)

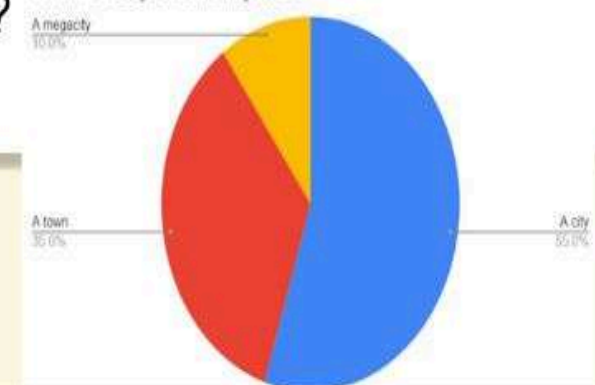
1.How old are you?



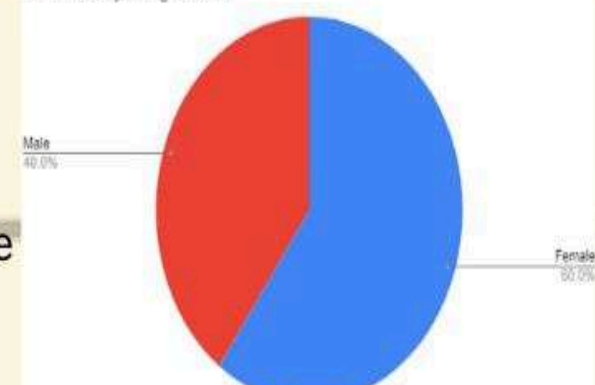
2.Do you love plants or gardening?



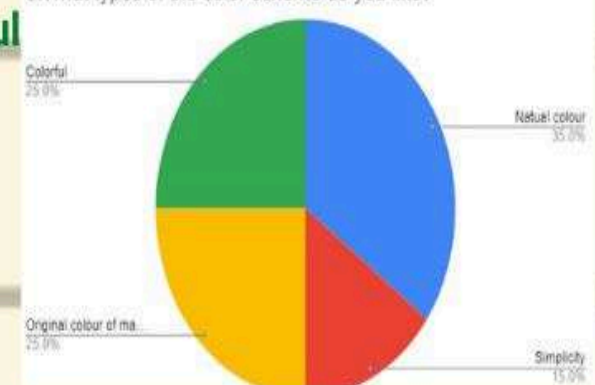
3.Where do you currently live?



4.What is your gender?



5.What types of the color scheme do you like?



6.What style of shelf do you like?

A.minimalist **B.Geometrical**
C. Modern **D. Natural form**

//The natural form is popular, which makes my
design should combine natural elements.

7.Do you want the shelf is ... the most.

A.Practical **B.Decorative**

C. Multi-functional **D. Fruitful**

//the multi-functional property and fruitful are
both the important points that people focus on.

8.What types of shelf do you like the
most?

A.Flood-based **B.Desktop** **C.Hanging type**

//The desktop type is popular (smaller in size)

9.What would you like to grow on the
shelf?

A.Vegetables **B.Flowers** **C.Herbals** **D. Others**

//Vegetables are the main part that most of the
people are focused, so the shelf should adapt the
vegetables' feature(i.e. root length, water level or the
need for creepers' racks)

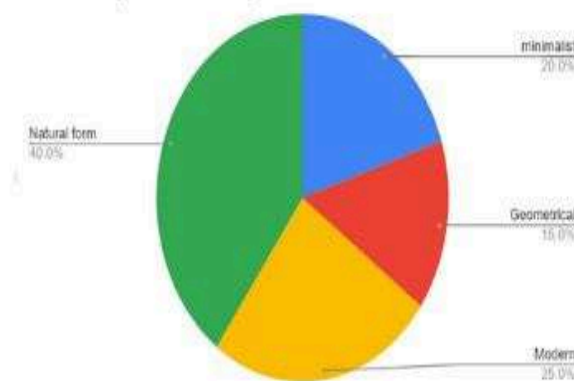
10.What's your ideal price for the shelf?

A. no more than 500 Baht **B.500-1000 Baht**
C. 1000-3000Baht **D. 3000-5000Baht**

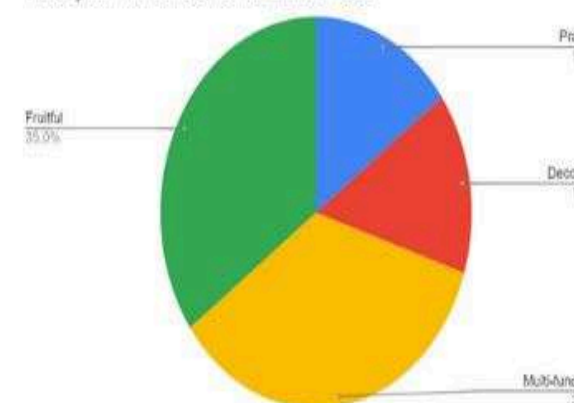
//people would like to have a balance of price so the
material of the shelf should not be too simple.

*20 people replied to my survey
for 2 weeks, which I learnt
many helpful suggestions.*

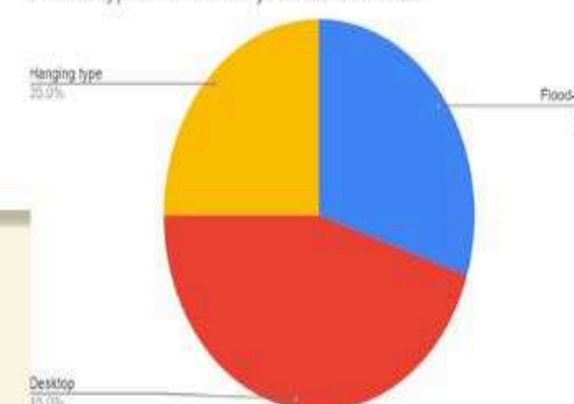
6.What style of shelf do you like?



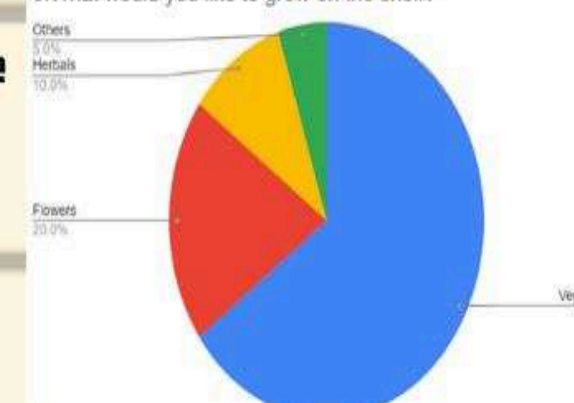
7.Do you want the shelf is ... the most.



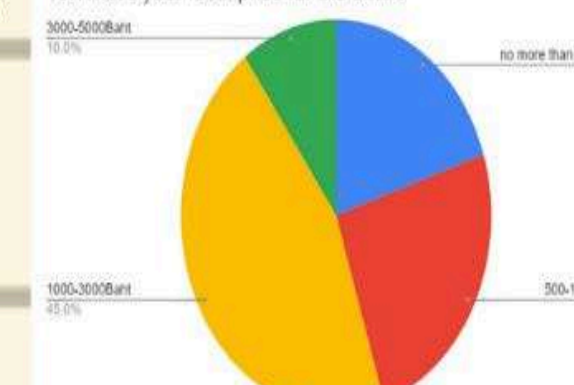
8.What types of shelf do you like the most?



9.What would you like to grow on the shelf?



10.What's your ideal price for the shelf?



THE THINGS THAT I
MUST CONSIDER:

- 1.mature shape
- 2.female customers take the
greatest number of market
- 3.nature-related shapes and
colours
- 4.multifunctional&fruitful
- 5.desktop(smaller size)
- 6.vegetable adaptive features

CONSULTATION

During the process of gathering ideas, I interviewed possible clients, who provided me with some other aspects of the project. Some ideas even make me excited because I never had them before. It is a brain-storm process, which also gave me opportunities to think about my project more deeply.



Clients 1&2



Overall, a suitable way of growing green plants in cities is as important as the climate change that we are dealing with now. Speaking back to the product, they love nature's colour, such as green for trees, yellow for sun and blue for the sky. It is also the colour scheme that I am aiming for after receives surveys.

"Although I like the appearance of wood, I still believe the combination of metal and plastic will make the shelf more sturdy and outstanding, which also limited the rate of deforestation." the miss said kindly. 'One thing you should be aware of is the water-resistance of the material.' They mentioned the property of the material a few times during our conversation. I believe plastic has better water-resistance property compare with wood and metal, so it gives me basic ideas of the shelf's structure.

The safety of the material is another thing they brought up. I need to do more investigation on this aspect later in my material research. "We don't want the shape of the shelf too complicated or full of sharp corners. It can be dangerous if the family has children." Then gave me another point that I should be aware of.

For the function, what they really want is a place to put gardening tools and appliances. So I should bring this need later to my design specification. They also have pet fishes at home. "If the shelf can also hold a fish pot, it will be nice since the nutrient solution is close at hand. Like what is in your design brief--combine hydroponic and aquaponics." after the sir read my design brief, he showed his pet fish to me. The fish pot is not big so I will consider this function.

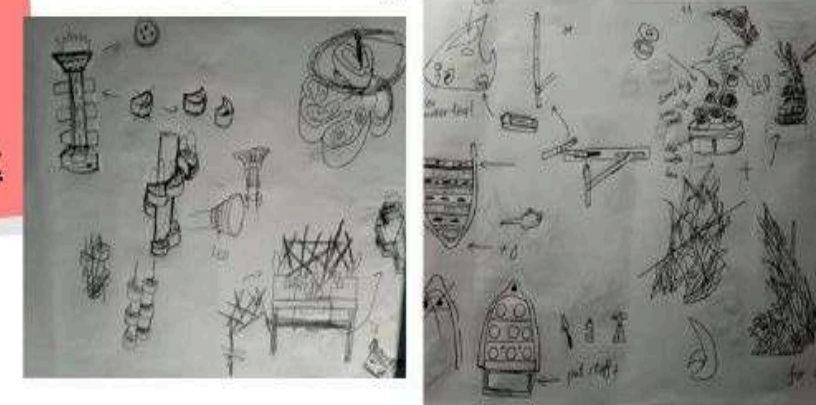
I interviewed students in my school, who are interesting in growing plants at home. They told me excitedly how many plants they have got. I asked if they get a shelf at home for plants. They all told me they don't have a specific plant shelf, but left plants next to windows, on the table etc. When I brought up the idea of hydroponic, they supported me, willing to providing ideas of what they want the shelf like.

"Thinking out of the box! You can create a shelf with the shape of the leaf," one of the students told me. I investigate the possible materials and tools while gathering ideas for the design. At the end of the researching, (later on) we had gotten more than 10 ideas. Combining the inspiration from the existing product, we have got the ideas of shape: leaf-like, bird-like, house-like, fish-like, tree-like; ideas of colour: green with white, blue with black, yellow with green; ideas of flower pots: recycling plastic pots, metal wires pots, 3D printed pots and normal plastic pot for hydroponic only.

Clients 3&4&5



By considering sustainability, we are more prefer the recycling idea for the flower pot. Based on my research on environmental issues, the system needs some typical electrical supports. One student told me we can design a self generator system for the shelf itself to generate the electricity it needs(renewable energy). Another boy told me I need a plants light otherwise the plants might die in a dark flat in the city. I love those ideas, which I might include later on.



"We really like the idea of the hydroponic shelf. It could bring cities to a new environmental friendly place. We live in the capital of a province, so there is limited space for gardening. The shelf can solve this problem." both of them said. "My gardening dream will come true." said the miss. "I don't need to squeeze in the shop every morning if I have that type of vegetable shelf." said the sir honestly. After sharing my idea, they felt proud of my 'ambition'. We talk through the environment, colours, possible materials and functions.

They told me there was a huge garden in their back yard two years ago. Because of the increasing number of vehicles, the back yard was dug and covered by cement for cars' parking. "Nature is moving away from us. We have already done too much." They signed. This story makes my goal clearer--solving the decline of the number of plants in cities. The couple shared more stories afterwards, including the garbage classification and people's choice of transportation. Air pollution is another thing they worried about. "The pollution from cars and factories is more like a daily normal now."

Hydroponic systems

Hydroponic is a way of growing plants without water. Each of the systems has its advantages and disadvantages. They all have basic structures and pros & cons. (red = all have, blue = some have)

structure: 1. **reservoir** 2. **plastic cup** 3. **medium** (e.g. coconut fibre, vermiculite, clay pellets) 4. some use pumpers/timers 5. fish pot (or combine with reservoir) as an aquaponic system

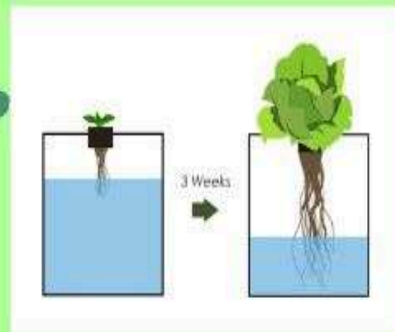
pro: 1. use **less water** than traditional farming techniques. (can be used in the desert) 2. **space saving** 3. no effects from weed or pests 4. can grow crops all years around (reduce import = reduce CO2 = fight against climate change)

cons: 1. some **investment** required 2. need for **technical skills** to control the system 3. some crops are not suitable to grow in this system (e.g. potato, carrots)

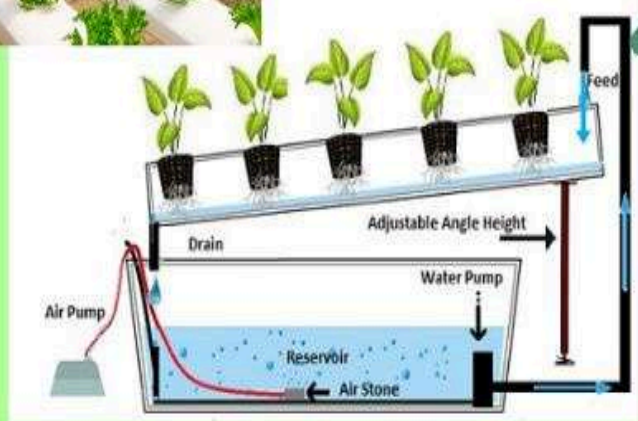
Wick

The ropes are used to absorb water and nutrition for the plant in pots.

- uuuu
- size limitation
- the absorption of water and nutrient might not even



Nutrient Film Technique

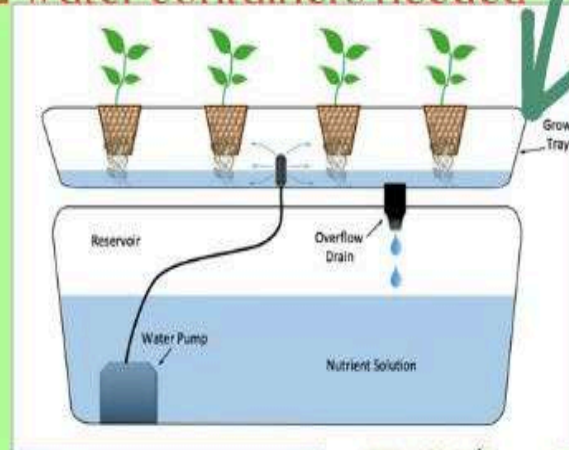


It is a system with flowing water so the nutrition is recycled and reused, flowing from the stream.

- no medium required
- plants grow faster because the oxygen in the root is enough
- continuous cycle, which is easy to maintain
- limitation of the size of the plant
- difficult to setup

Ebb and Flow

2 water containers needed

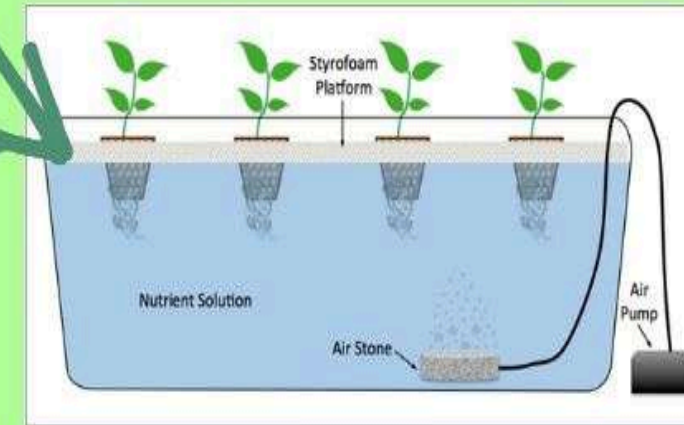


It works by flooding and draining the roots of the plant which simulates the situation of plants in nature.

- fast growth
- easy to setup
- can use in Aquaponic
- unstable pH
- some plants cannot fit in this system



Deepwater Culture



Put the plants' root entirely in the water.

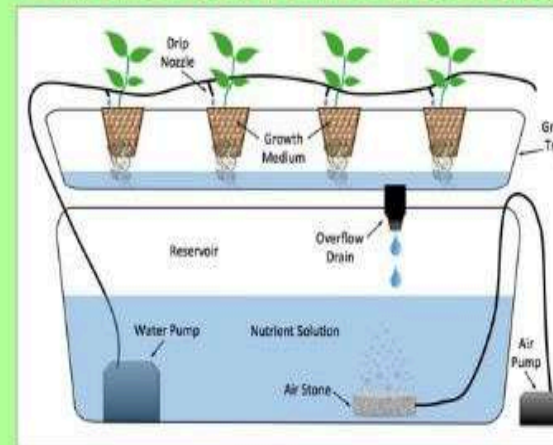
- easy to set up and operate
- plants grow faster because less water and nutrient required
- algae could grow into the bucket--regular clean required
- air pumper must function properly to oxygenate the nutrient solution

Drip System

Dripping nozzles drip the solution to plants, which flow back to the reservoir.

- providing more oxygen to plants
- less water and nutrient required
- need to clean the grow tray regularly
- pumper must function properly
- an extra water pumper required

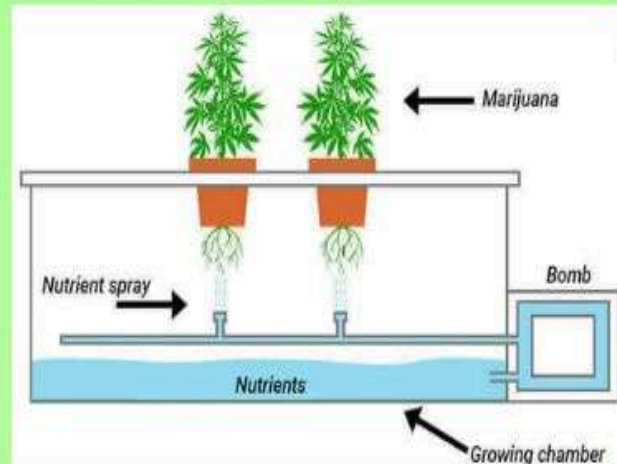
2 water containers needed



Aeroponics

Spays spray nutrient to plants' roots, which doesn't touch the water so it is easy to control.

- no need for a gas pumper
- no growth medium required
- less water and nutrient required
- need to clean the root pots for plants
- pumper must function properly



Environmental problems and how does hydroponic can help us:



I visited the local hydroponic farm during my investigation. I met some workers there and discussed the characteristics and future development of hydroponic. They told me as the environment gets worse, hydroponic will become more popular, which can even solve the problems that we are facing today:

- 1. climate crisis
- 2. food wastage
- 3. air pollution
- 4. plastic waste
- 5. deforestation

They then explain all the details about this 'future saver' planting system.

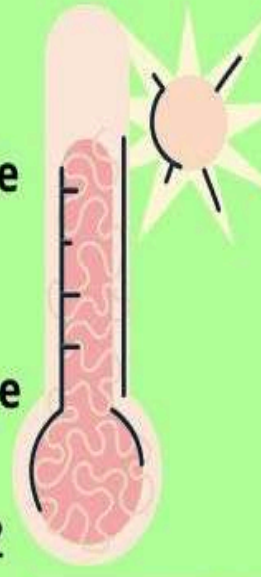


1. climate change:--global warming

It can be related to all the problems that we are facing nowadays. If there are fewer trees, more greenhouse gases will stay in the air, aggravating global warming. If there is too much air pollution, greenhouse gases will increase unavoidably. If there are more plastic waste, indicating more crude oil and non-renewable energy is used, leading to high carbon emission. If there are more food are be waste, it means more methane gas will be produced during the decaying process, creating more greenhouse gases. But with a simple hydroponic system, people can grow plants more easily, purifying the air and saving spaces at the same time, fighting against climate change in many ways.

2. food wastage

People waste food for many reasons. In most of the case, they don't really care. 20 per cent of food are wasted each year. However, if the hydroponic system introduced in each house, people will have conscious of wasting since the food grows from their own house and cost their own food.



3. air-pollution

Green plants can do photosynthesis to absorb carbon dioxide. If we have more plants on the Earth, using hydroponic as a base, the number of green plants will increase exponentially, purifying the air around us more efficiently, especially in cities.



a picture of the Meina river that I took from Bangkok



4. plastic waste

Although plants can't dealing with plastic directly, they still can fix the waste gases which are produced by burning plastic or making plastic, in some way, dealing with air pollution from plastics

5. deforestation

Trees are cut down to grow crops or build as cuttle farms. it causes severe problems to the Earth's climate. If the hydroponic becomes popular, we don't need to cut down extra trees for planting crops. Each of us can grow plants at home, in school, in working areas, limiting the need for farmlands.

minimalism



MOOD BOARD



GREEN LIFE

I like its outside rack--like a house or a basket full of vegetables, even though it only has less space for planting.



An upcycling shelf with a dripping system. A beautiful design but it has to be constructed carefully with a lot of thinking and cares.



It uses the nutrient film technique, which is simple in design. The turning tower plants is creative, but it is not very sturdy.



A hanging style shelf with a simple design and colour. It looks strong but it only has limited space for plants

PRODUCT
IMAGE
BOARD

Existed Products Analysis

By analyzing these products, I get to know their positive and negative factors. Making me more likely to get my conclusion in the design specification.



plastic type:PP

Automatic Hydroponic desktop planting box

Aesthetics: This is a neat and simple desktop hydroponic box. The beige colour box gives a feeling of soil, which is in sharp contrast to plants at the top. Its shape is ordinary, but minimalists may love it.

Client: This product is easy to construct, which also gives people a feeling of being close to nature. So I reckon it is more suitable for the middle-aged group(30+) or minimalists based on its colour and style.



Ergonomic: As the picture shows, it has handles on both sides for lifting. The water level gauge is tiny but useful, which highlights the designer's attentiveness.

Functions: This design follows Deep Water Culture. Because of its size, limited the species of plants customers could plant, which is also not a wonderful choice for a large family. Also, customers have to pay extra money to grow light otherwise to place them under sunlight(shorten plastic life). Despite the plants' normal functions, the product itself doesn't have any highlights. It is easy to clean however since it has generally fewer components.

PRICE:609.69THB

Cost: The price is nice and affordable. It offers all the components for the system and the plastic itself is not expensive.

Size: The product is relatively small, which only has 6 holes for planting. So it cannot produce much food. It can be put anywhere since it takes less space.

Safety: It doesn't have sharp corners so it is safe for children. The only thing that needs to concern about is the loose lid above the reservoir, which is easy to be overturned, causing water flows out.

Materials: Plastic is the only materials on this product. It gives a property of lightweight so it is easy to carry around. However, the plastic will harden slowly under the sunlight and broken eventually. So it doesn't have longer sustainability. The plastic itself is harmful to the environment, even though it uses eco-friendly plastic.

Climbing Trellis Super Indoor Hydroponics Growing System

plastic type:ABS



PRICE:2771.34THB

Aesthetics: This product shares a similar appearance as ladders. The overall colour style is boring but it fits with green perfectly if creepers are planted. It also follows austere minimalism. It is a floor-based shelf, and the box is a simple design, which really suitable for trellis at the top.

Client: The product is eye-catching and has its own characteristics, so people who want growing various vegetables are target customers. Base on its price, it is appropriate for the middle class, who are more focused on the productivity of food but not the decoration of the home.



Cost: The price is higher than the basic hydroponic shelves because the system is better. If we think about productivity, the price is reasonable after all and plastic does not cost too much.

Size: The product is simple and place-saving. It provides more space for plants to grow although it only has three planting areas.

Safety: It is a high shelf so it is important to make sure each part are assembled correctly and sturdily together, decreasing the possibility of fall down. The bottom part is closed tightly, which prevents the water from spilling if it is accidentally knocked over.

Ergonomic: This shelf has long poles, which share a similar height to ours. So it is easy for picking. The only short is its portability that the shelf is hard to move once plants are grown on a trellis.

Materials: Plastic is the only materials that are used.

Generally, plastic is water resistance and stiff. On the other hand, it is a cheap and light material although it is easy to break after a long period. The trellis in particular should be aware because it is tall, which could crack under too much sunshine or after being used for ages.

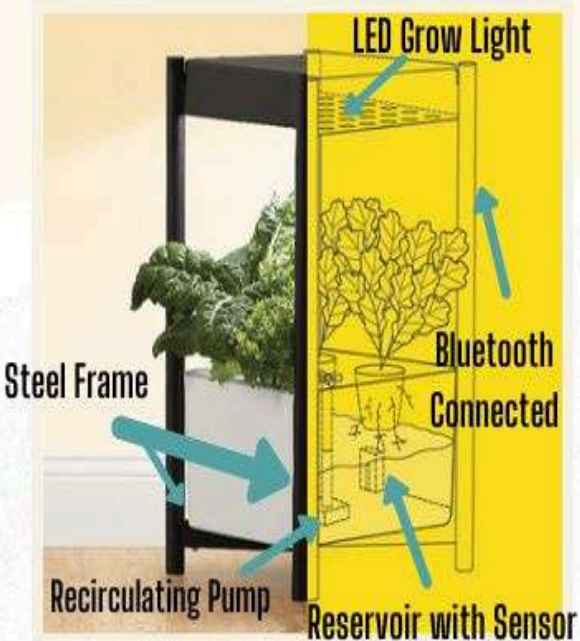
Functions: This design follows Edd and Flow. It is easy to assemble and trellis can be removed if the customers don't need them. Because of its form, the main purpose of the product is growing creeper vegetables or flowers. It only has three places for planting, so growing creeper vegetables is better than growing lettuces to increase production. Generally, It is very decorative, which provide green colour at home. It has few components so washing is not a big deal.

Miracle-Gro Twelve Indoor Growing System

Aesthetics: This product is larger, which looks modern with the black and white colour scheme. Its shape is a cuboid, which is able to fit in other furniture at home easily. Green and white give a sense of freshness of this project, making it not boring but refined.

Client: The product can be used as a table, so people who love multi-functional product should love this. It is more suitable for people who pursue a high quality of life. Because it is a little pricey, I believe people who have financial bases will buy it.

Cost: The price is high because of its multi-functions properties (use as a side table, can be stacked up as a whole), its size and its steel legs.



Ergonomic: As convenient as sitting at home, it contains simple components to assemble a smart Bluetooth controlling system and sensor of water level in a reservoir. It has the same height as normal side tables

Size: The product is as big as a side table so it is bigger than usual hydroponic systems. Although it takes more space, it is useful since the surface of it can be used to place appliances.

Safety: It has sharp corners, which could be dangerous for children. The steel structure is heavy but strong, so it is important to pay attention to it when we move the shelf. Apart from this, the use of electricity should be considered too.

Materials: This shelf uses steel and plastic as its main materials. The steel-made frame as the outer part can bear much weight, which makes it works also as a table. The plastic base and top are light in weight, decreasing the total weight of the project. The designer uses both materials' properties to create this project.

Functions: This design is a drip system. It has different functions such as supporting stuff and illuminating the room. Although it only has 4 holes for planting ranges of vegetable, it still provides enough space for each plant. The LED light is connected with a sensor so you can distinguish the state of water by flashing light. It is also decorative furniture at home, which can create a tranquil surrounding.



PRICE: 7256.33 THB

Herbert

Aesthetics: This product is a combination of nature and modernity. It has a wall hanging structure with a simple appearance. The wood and plastic create a lovely and mature feeling. The overall shape is a rectangle and the wood shows a sense of classic. Organic plants with wood frame are the best decoration at home.

Client: This product is the best example of modern design. It has a high rate of harvesting and various options for plants growing. Because of the use of new technology and its special design, the customer must have a high income and enough financial supports.

Size: It takes no space on the ground and less space on the walls, which also fit with walls perfectly.

Cost: The price is higher compare with other hydroponics. Wood and its system take most of the cost in the manufacture and appearance design are one of the reasons that make it expensive.

Ergonomic: This product can be hang up based on clients' height. The length and width are reasonable. The LED light can be control by phone, and the system is intelligent.

PRICE: 9425.33 THB

Safety: Because it is a hanging type, it may cause hidden dangers when children play under it. It has some sharp corners and the shelf should be hang up properly in case of dropping.

Materials: Wood is the outer frame of this product, which gives it a natural feeling. However, it increases the weight of the project. The plastic is used to make pots and frame for the LED light. They are stiff and light so they are great options.

Functions: This design is a Nutrient Film Technique. It has complex assembly procedures. The shelf is not easy to move once it has been hung on the wall, but it is undeniable productive--15 plants can grow at the same time! It is an aesthetic and practical shelf with smart technology. It also expands the choose of vegetables, decreases food waste and saves money.



HARDWOODS:

Meranti AVERAGE DRIED WEIGHT:500-650 KG/M3



The wood shows various colour from light brown to dark red, with straight grains. It is easy to work and strong. Although it is not resistance to the insect, it doesn't have specific allergies, which is safe for everyone. The cost of wood is reasonable.

pros:1.dulling effect 2. resistance to warp or twist

cons:1. does not resistant to insects 2.need maintenance(e.g. put on vanish)

Highly available in Thailand

Chengal

AVERAGE DRIED WEIGHT:915-980KG/M3



The light yellow-brown colour will darken when it is exposed to the sun. It has interlocked grain, which is shallow. The wood itself is resistant to termites. Because of its dense and strong property, it normally is used to make boats, beams or heavy-duty furniture. It also used in decking.

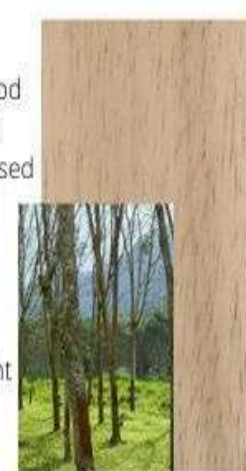
pros:1. highly durable 2. withstand harsh weather 3.need little maintainance

cons: 1.expensive 2. the wooden dust might cause occupational asthma

Just expensive

Rubber wood

AVERAGE DRIED WEIGHT:560-640KG/M3



It has blond in colour, which is darkened over time. It has a straight grain. It's easy to work but the perishable and little resistance to decay becomes a drawback of this wood. Commonly, people use it to make furniture or kitchen woodware. Someone might find it allergic.

pros:1.cheap 2. easy to work

cons: has the tendency of warping while it is drying

Decaying

SOFTWOOD:

FIR AVERAGE DRIED WEIGHT:415KG/M3



The firs usually grow in the North part of the global or the cold mountain ranges. It is a utility wood. People commonly use it as construction lumber or flooring. The good thing is it is SUPER easy to work. Straight grain and pale colour make it looks great. Resistant to insects and decay, it can cause sneezing. It is fairly strong compare with other softwood.

pros:1.looks nice 2. easy to work

cons: high cost for shipping and wood itself

Not highly available in Thailand

SPRUCE AVERAGE DRIED WEIGHT:405KG/M3



The white and creamy colour with straight grains gives a peaceful feeling to people. Comparing with the firs, they have similar properties: easy to work. It is cheap so we commonly make papers, constriction lumbers, and crates with it. It has a medium weight, which is also sturdy.

pros:1.cheap2.strong

cons:1.low resistance to the weather

Highly available in Thailand

PINE(Pinus Kesiya)

AVERAGE DRIED WEIGHT:610KG/M3



It is one of the most distributed woods in Asia. It is tall with straight and obvious grains. Its light colour. Also, the wood is resistant to the swelling and easy to work. It normally is used as timbers for construction, furniture or veneers. It is native to Thailand, which is cheaper and easy to find.

pros:1.easy to work 2.lightweight 3.

cons: easy to leave scratches or another inside

Highly available in Thailand

ENGINEERED WOOD:

HARDBOARD DENSITY: 500 KG/M 3



Also called High-density fiberboard(HDF), a type of fiberboard is made by sawmill or forestry remainders. Compared with MDF or particleboard, it is denser, which can be used in furniture or construction. However, the wood itself is not waterproof that it needs to cover with a waterproof coating. Productions of the hardboard have two ways: wet or dry. The wet process makes one side smooth and one side textured; the dry process makes both sides smooth.

pros: 1. affordable 2.denser

cons: not waterproof

It is not a good idea to build the whole furniture with it.

MDF DENSITY: 500-1000 KG/M 3



This manufactured wood is well-know as medium-density fiberboard. It is made of wood residuals combined with wax and resin, which means it may swell or expand in a wet environment. So the MDF usually are used for indoor projects. The wood is not harmful to the environment, but the resin in the making process can bring a type of compound called 'Formaldehyde'. This compound could pose health risks at the area with high concentration.

pros: 1.generally denser than plywood, but not stronger than plywood 2. It is easy to glue, shape and keeps its dimensions

cons:1. not waterproof2.health concerning material

Despite this, a veneered MDF is beautiful with amazing 'grain'.

PLYWOOD DENSITY: 460-680KG/M 3



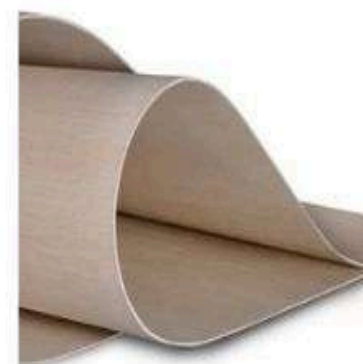
Plywood is a wood made with resin and wood veneers, which has different types, depending on the types of wood residuals such as the hardwood residuals will make hardwood plywood.

pros:1. reduce the possibility of split 2.not easy to expand or shrink 3.have the same strength on the surfaces. 4.Compare with MDF, it is stronger. 5.It is used in high-quality applications for either outdoor or indoor like floors, furniture, or wind turbine blades.

cons: moisture environment still can affect the strength of the wood.

Overall, it is idle wood in many cases.

FLEXIBLE PLYWOOD



It is still made of many thin layers of wood glueing together. The thickness is various between 3mm to 12mm

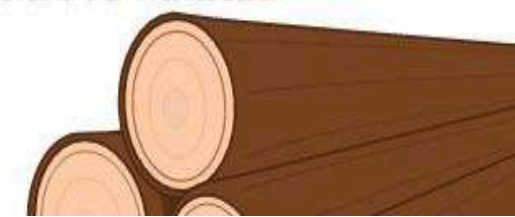
pros:

1.flexible plywood can be used to form curves without broken.

2.save time for construction

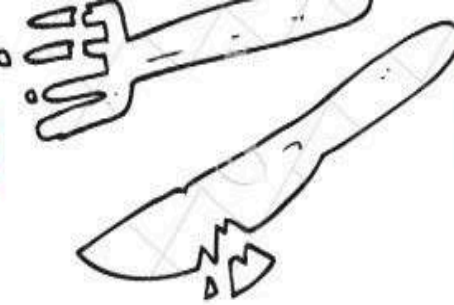
cons:

1.Not durable like plywood.2.using for furniture rather than in construction



I VISITED THE LOCAL TIMBER SHOP AND FURNITURE STORE, INVESTIGATING THE EXISTING WOOD'S PROPERTIES RELATED TO THEIR SUSTAINABILITY. OVERALL, THE ENGINEERED WOOD IS THE MOST RECOMMENDED DUE TO ITS SUSTAINABILITY.

Plastic research:



RECYCLING CENTER



PET-polyethylene terephthalate thermoplastic



PET is usually been used as materials for water bottles or food packaging. It is transparent and light in weight. Because it's a thermoplastic polymer, the property of stiffness and ductile makes it a great packaging plastic. They are 100%recyclable(oil-based polymer), which can be converted to plastic fibre afterwards. It resists high temperature as well, which is also can be used to make trays for microwave. The way of forming is very wide: Vacuum forming, injection moulding, compression moulding and blow moulding.

PMMA-methyl methacrylate(acrylic) thermoplastic



High transparency is the main properties of this material, so it can be a good alternative material for windows that don't need to consider impacts. It forms in many semi-finished products such as sheets, tubes or rods. It is brittle, which cannot bear a strong impact. The chemical resistance properties still make it commonly being used. It is hard and light in weight, which carries properties that most plastic has. Moreover, it is biocompatible so it can be used in medical. Don't forget it is 100% recyclable.

Quite a lot of plastic that I collected. I saved some for upcycling later.

My aim is using recyclable plastic materials in my project(or perhaps upcycling). In order to find more information, I visited the local recycling centre. By the way, I recycled all the PET bottles that I had accumulated over 4 months. It is a huge place with mountains of different garbages, mainly inorganic. The workers listed out some of the recyclable plastic to me and explained the process of recycling. Some of the plastic needs a lot of procedures, which also emitted greenhouse gases during the process. So eventually I have a clear idea of my choice of plastic material.

HDPE-high-density polyethylene thermoplastic



Similar to PET, HDPE is used to make containers for liquid substances. It is a much stronger option compare to PET, which is also be used to make floor tiles or plastic lumber. It has more colour choices and the product can be either translucent or opaque. It is 100%recyclable as well(oil-based polymer). The material itself is non-toxic, so it is safe to make children's toys. We can only use injection moulding for it. Injection moulding is recommended.

UF-urea formaldehyde thermoset



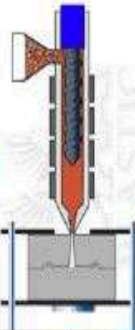
This material comes from urea and formaldehyde. It can be used either as resin or polymer. After it is heated, it shrinks, which requires a higher temperature. The high tensile strength makes a hard and strong surface. However, if it is broken, the fissure will expand and maybe eventually break. It has low water absorption as well so it is popular in the wood industry--the adhesive of the pressed wood board. It releases less formaldehyde than other thermosetting plastic. Also, it could be completely recycled.

injection moulding



I am in the school's ECO Club as well, which is another chance for me to study plastic recycling. It is a shame that every time when I visited beaches, the most 'attractive' sight is no beautiful waves but 'piles' of garbage. I always collect them but I know my own strength is too little. However, it is great if I collected plastic garbages, especially type 2,4 and 5. I can literally clean and bring them to school since they are recyclable in school. In the school's ECO Club, we use injection moulding to recycle these plastics. Everyone in the school collects them and helps the school's ECO community to grow. It is a creative and 'hot' process, which I will explain below:

- 1.Clean the plastic with water and dry them after. It is important to make sure it doesn't contain anything that is not plastic or not the same types of plastic.
2. Shred the plastic using a shredder. We can separate the colour of plastic, but it is special in multiple colours.
- 3.Melt the plastic in the injection machine, which needs to be preheated to 180-190 °C. Put the plastic shred into the machine and leave them a while for them to melt. At the same time, put a mould under the nozzle. The mould should be made of metal which can be cut off by CNC machines. After plastics are completely melted, pressing the reciprocating screw down and wait for the melted plastic fill the mould. Then the will plastic form the desired shape inside.
!!!The one on the right is a sea turtle:)



PLA-polylactic acid thermoplastic



PLA is made from renewable resources, unless other oil-based plastic. Hence, it is biodegradable, which makes it stands out(made from lactic acid). The properties of the plastic including its high flexibility and resistance to high tension. It has a generally low melting point, which makes it easier to melt and used as 3D modelling materials. It also can be used in healthcare, the industry of food packaging and cosmetics, especially for some companies who want to make their product more sustainable.

Epoxy resin thermoset



It is a versatile resin that can be used for many purposes(adhesives, sealants, coating etc.). Its functional group epoxide, which results in many types of epoxy resin with different molecular structure. Moreover, it can combine with multiple curing agents, performing different functions. It offers high strength if it is used as adhesive. Also, it resists chemicals, electricity and moisture. The good flexural strength makes it better than polyester resin. When it is used as additives, it can increase flexibility, decreases flammability and increase fracture toughness.

METAL RESEARCH:

NON-FERROUS METALS

they all have those general properties

PEWTER (MELTING POINT: 170-230°C)

It is a tin-based alloy, with a long history as iron. It is an alloy with 91 percent tin, 7.5 percent antimony, and 1.5 percent copper. It gives a bluish-white colour with a shiny finish, which resists tarnishing. It is normally used to make dishes, decorative items or ornaments. It is a recyclable metal as well.

ALUMINIUM (MELTING POINT: 660°C)

Light material with a weight of 2.7 g/cm³, it is super ductile and increases tensile strength as temperatures drop. It forms an oxide layer, which makes it resist corrosion. The surface reflects light very well, and it is 100% recyclable. It is not toxic so it is used to make cooking utensils. Bicycles, window frames and ladders are always made by it because it is light. It is really cheap since it is one of the most abundant metal too.

COPPER (MELTING POINT: 1080°C)

As the oldest material that human has used, the copper is the most versatile material in workshops. It is ductile and malleable with a red-brown colour. It resists corrosion from water, which is useful as a pipe. Cookwares and electric lines are normally using copper as the materials. The copper is 100% recyclable.

The metals have 2 major types:

- **ferrous metals** (The metal contains iron)

general properties: **high melting point, very hard, conduct heat and electricity, react with oxygen so they need to be coated before applying to my shelf. Ductile and malleable.**

- **non-ferrous metals** (The metal doesn't contain iron)

general properties: **have colour, generally low melting point, conduct heat and electricity, less reactive with air or water.**

FERROUS METALS

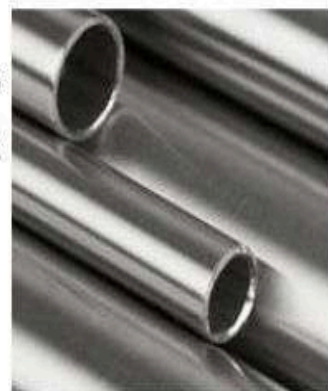
STAINLESS STEEL (MELTING POINT: 1400°C)

Famous for its corrosion resistance and heat resistance, stainless steel is an alloy with iron, carbon, and others (Cr, Ni, Mg, Mo, N). It is a strong material with good durability. The great malleability, low maintenance requirement, and appealing appearances make it even more popular. Apart from that, it is also environmental friendly.

It is often used to make the kitchen equipment.

CARBON STEEL (MELTING POINT: 1456-1800°C)

It can be classified into 3 categories: low-carbon steel, medium-carbon steel and high-carbon steel. The more carbon it contains, the tougher it is (low ductility). Low-carbon steel (<0.25% carbon) has good weldability and low cost, which can be used as food cans. Medium-carbon steel (<0.6%) resists wearing, using as railway tracks. High-carbon steel (<1.25%) is the toughest one, which often is used to make cutting tools or screwdrivers. All types of carbon steel are eco friendly.



WORK HARDENING - ANNEALING

Annealing is a heating process that heats the metal to a certain temperature and leaves it to cool down slowly. This process makes the metal become soft so they are easy to be cut and shape. The metal aluminium needs to take care of while heating, otherwise it might melt.

Work hardening in the contrast is a process of heating metal as well but put the metal into the cold water quickly after being heated to a specific temperature (colour). It is heated again with great care because the colours change will happen rapidly. At last, put the metal on another conductive metal piece, making it cools down slowly. The hardened metal could be either a desirable feature or not. Such as the aluminium can become very brittle, which is easy to split.

TOOLS & MACHINES

I will learn and try to apply all of them in my manufacturing and testing process

MEASURING TOOLS



steel ruler

use to measure and marking



try square

use to mark out 90-degree lines on edges of materials to check if it is a right angle.



marking gauge

use to measure two parallel lines on an edge of a wood, normally for marking out joints



odd-leg callipers

use to scribe out a line at a certain distance from a materials edge



micrometre

measure the very short distance of a working piece (e.g. the thickness of a metal sheet)

SHAPING TOOLS/MACHINES



tenon saw

use to cut the wood, especially the joint



band saw

use to cut wood and plastic in either straight line or curve



pillar drill

use to drill accurate holes on wood, plastic or metal



electric drill

use to drill different types of holes depends on the drill bits



guillotine cutter

use to cut thin working pieces. (e.g. metal sheet)

FORMING TOOLS/MACHINES



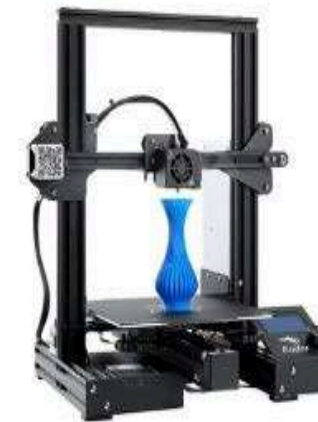
laser cutter

use to cut the materials in a precise size, normally used in the industry



strip heater

use to heat a thin plastic sheet, which makes it easy to bend



3D printer

use to producing 3D objects with accurate shapes, which can be either metal or plastic



vacuum former

use to reform plastic sheets to different hollow shape, with a mould



injection machine

use to create many objects by the melted plastic and inject them into a mould

Finger Joint

Finger joint is a simple joint but firm joint that including these basic steps to make:

1. Marking out cut lines on the flat and smooth timber

I used basic tools to measure perfect cutting lines. In order to get the exact cutting area, I put two edges of wood together to mark out the fine cutting line.



My joint's width and length = 3 cm : 4 cm

The width of the joint depends on the thickness of the wood. The length of the joint should fit the chisel's width to make it easy to cut later.



THIS IS A GREAT WAY TO KEEP THE JOINT AREA IN THE SAME SIZES.

2. Cutting the joint

We all say '3 years to measure and 3 days to cut.' when we have marked out any wood crafts. SAME HERE! I chose a rubber hammer and chisels to complete the cutting process:



THIS IS A METHOD WHICH IS EASY TO CONTROL AND IT DECREASES THE ERROR RATE a rubber hammer (give chisel a force to break through the surface of the timber)

a bench hook a chisel (for removing wood)

(hold the timber's position) !!!hold it straight

Advantages:

1. Easier to make
2. Looks smooth and beautiful
3. Sturdy rate: 65%
4. Less waste

Disadvantages:

1. Time-consuming :(
2. Not SUPER strong
3. Difficult to make all the parts really accurate

USE IN: FLOORBOARD, ORNAMENTS...

IT IS PRETTY WHEN IT HAS A SMOOTH SURFACE

Conclusion:

The finger joint is a stiff and common joint, although it is easy to break when we cut or trim it. I don't suggest to use these types of joint if you are doing a big project.

3. Smoothing the joint's surfaces

I tried the **sanding machine, abrasive paper and rasp**. At first, I used paper and found out it is too weak to smooth larger area of uneven bulges.

So I changed it to rasps. They are stronger than abrasive paper and able to smooth big parts.

After the smooth inner surface of the joint, I connected them together and smoothed them again on the sandy machine. The reason why is I find it useful to make the connecting part of the joint flatter and smoother, which also gives the timber a good surface texture.

Dowel joint is a strong joint, which takes generally less time to make:

1. Marking out the drizzle position

Firstly, I drew a line along the edges of two timbers. It gave me the area where I would drill holes.

Then, I drew 3 vertical middle lines to separate the drizzling surface. At last, I drew a horizontal line intersected with those vertical lines to get 3 accurate middle drizzling points

Advantages:

1. give the product circled pattern
2. Sturdy rate: 90%
3. Less waste

Disadvantages:

1. Time-consuming :(
2. difficult to make all the parts really accurate



cutting off the extra dowels outside of the joint.

Conclusion:

Dowel joint is a firm joint. It can be used on any big project. However, it takes time to measure and drill the holes. It is still useful after all.



doesn't need so much strength. Remember to clamp it

2. Drill it

Drizzling is the main part of making a dowel joint. If there is anything wrong, the joint cannot fit with dowels, so the joint is not strong enough. I tried it with two methods: pillar drill and normal hand drills.

3. Add dowels with glue

With the accurate holes, I tried to put them together. However, the other way is glue two pieces of wood together use 'butt joint', and then drill holes. I dipped dowels into the glue and using a rubber hammer to put them in. After cutting the extra bit off, it was time to sand it.

Dowel Joint



Conclusion
The mitre joint is invisible after it is constructed. The shape of the joint remains a beautiful angle to users as if there is no joint at all, although it is not as sturdy as the dowel joint.

USE IN:
CHAIR JOINT OR OIL PAINT FRAME

the joint is very common and easy to construct



USE IN:
BOXES OR BEAMS ON THE ROOF.

THEY BOTH USE PVA TO GLUE



1. Marking out

First, I did the same thing to get a straight line on one of the wood. Then I measured out 45 degrees on each corner by the set square and splitting the corner of each two-piece of boards.

2. Smoothing the edge

Smoothing the wood board on the sanding machine. It is an important yet difficult thing to make the corner flat and perfect.

3. Glue each part together

Finish the joint by glueing two pieces of joint together. It must flat clean enough to allow the corner to join and stick together, otherwise, a gap will form in between. Luckily, my joint is work, which is fitted together.

Advantages:

1. Easier to make
2. smooth surface
3. Sturdy rate: 80%

Disadvantages:

1. not too sturdy
2. need time for smoothing

Mitre joint

1. Check the height

Making the joint looks smooth and professional, it is important to measure the correct length of each board.

2. Glue two pieces together

using the PVA glue to glue two pieces of wood together to form a butt joint. There is nothing attached to them except the glue. Generally speaking, it is the least sturdy joints. However, it can be used with screws to decrease the rate of broking.



the joint is super easy to construct.

3. clamping



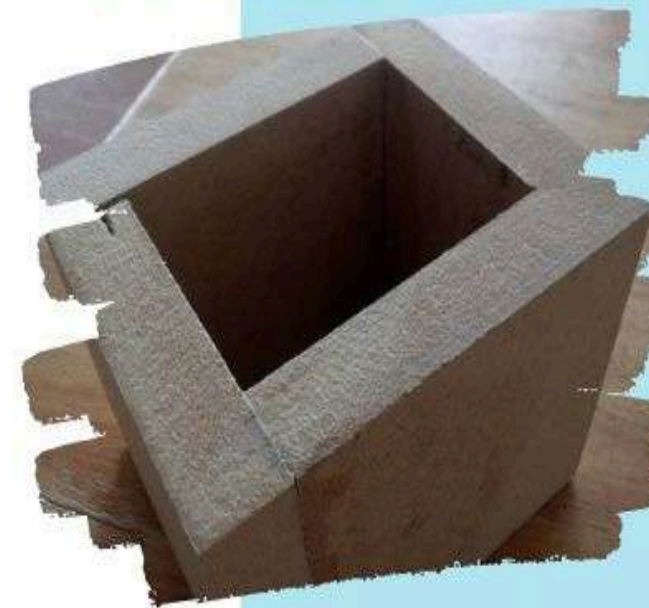
After putting on the glue, the joint needs to attached firmly. Hence, we need to use one or more G clamp to clamp the box together. Making sure both sides are on the same lever, we should clamp it with some scrap of material underneath to protect the surface of the box. After the glue is completely dry, the joint is finished.



Conclusion

The butt joint is the simplest but the weakest joint. It is really easy to construct, but it needs to clamp with great care.

OVERALL CONCLUSION:



After the primary research, I get to know how does some basic joint work and its construction details. There are more joints in reality, so I still need to do secondary research about it. These joints all have their own pros and cons. I personally prefer the dowel joint more because of its decorative factor and sturdy characteristics. But they all not the best choice for a shelf, which is clearly bigger than a box.

FURTHER RESEARCH

They all can be made with different materials depends on their uses

NAILS AND SCREWS:

FLATHEAD SCREW(WOOD)



This screw can make the assembling part invisible and flat. The head of it is various, which can be used with different bits. It is a strong and durable screw, the best choice for fastening metal sheet. Before assembling, the user needs to drill a pilot hole first and a countersunk hole after, which make it easy to screw in after.

COMMON NAIL



It is generally used in framing. Their shank is thick and their heads are wide. Because of its thickness, it is strong but it is likely to split wood.

BOX NAIL



It is like a common nail, but it is much thinner and suits the wood materials. It cannot be used in a structure project since it cannot bear too much strength. It is, however, less likely to split the wood.



The claw hammer can perform the process of put in nails and put out nails

WASHER



The washer is a small circular disc shape with a hole in the middle of it. It can be used with screws together to perform these functions: 1. distributing the load (if a screw is driving into wood, the washer could reduce the risk of cracking.) 2. increasing the space (if a screw is too long for the thickness of the material, the washer could make it not drive all the way through) 3. absorbing vibration (some washers are made by rubber to avoid the damage of the material from vibration) 4. preventing liquid (they can be used in pipes, which can block the waterway on the surface of the object.)

SOLAR PANEL AND PLANT LIGHT:

Monocrystalline is the most efficient one (20%), which is made from pure silicon. It is also the most expensive one on this list.

Polycrystalline is the second efficient one (15%-17%), which is made from melted silicon fragment crystal and it is a newly developed solar panel.



Monocrystalline Polycrystalline Thin Film

Thin-film is the cheapest type of panel with less efficiency (11%). It is extremely small and newly developed. It is made by lots of different materials, including silicon



FLORESCENT



HIGH-INTENSITY



LED GROW LIGHT

LIGHT (COMPACT/TUBE)

For CFL, it is convenient to use, which is a better choice for a small space. It is intense enough for both vegetables and plants to grow. Same as tubes, they all don't give too much heat, but the tube can't give intense light like CFL.

DISCHARGE (HID) LIGHT

It is an old type of grow light that people have been used for a long time. It is super intense, giving out a large amount of heat. So the electricity bill needs to be considered and it must put further from plants. However, it works for both vegetables and plants

The popular grow light nowadays, which everyone is affordable. It can last longer than other types of light and giving out less heat. It also gives out intense light, which even can penetrate through the leaves.

HINGES:

Based on the questionnaire, I think a drop leaf table is necessary for a shelf, which can be either put solar panels or the utensils for people to use. So I investigate these hinges and brackets:

DROP LEAF HINGE

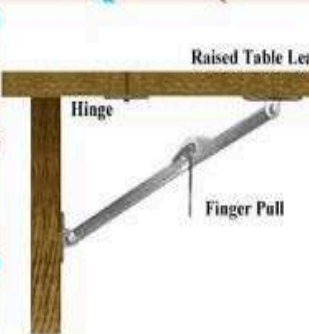
Like its name, a drop-leaf hinge is used for a drop leaf table. It requires some professional techniques to assemble it. The hinge itself needs to fit inside the table, so the working pieces' sides have to be routed accurately. The angles of the routed sides should be assembled a bit outward to make sure it is perfectly fitted to the hinge. When it is mortised into the pieces, a router is needed to remove a thin gap to let the jointed part of the hinge sink inside of the material. After it is assembled, the table can be lifted and dropped with the help of a supporter. It looks invisible to the users. The hinge supporter can be varied in appearances. They are all performing as a bracket to support a hinge. In the drop-leaf table, they are the supporters under the table and bearing weight from the table surface. It is not necessary to be a finger pull but also can be a wooden supporter, a bracket that can be put under the drop leaf table. In conclusion, it is time consuming and material wasting if I use both the hinge and the supporter.

FOLDING BRACKET

A folding bracket is a better choice for a leaf drop table. It requires only one type of component. However, the bracket needs more space for constructing, which means it is clearly visible on a shelf. It can also be dangerous since it closes with a strong force. If it is used to support a table, it could be super sturdy. The size of the bracket is also a problem that needs to be considered. They all appear as a large piece and quite heavy, which is not necessary for a simple plant shelf.

90-DEGREE FOLDING HINGE

It is a specially designed hinge that only can be folded 90 degrees. It is normally used on foldable table legs and drop leaf tables. It has an inter-locked function, which can lock the hinge when it is opened. The hinge is super strong after locked, which can bear the pressure very well. Although it appears a bit obvious on a shelf, it is generally smaller and lighter than a bracket, which requires a little space for assembling.



ADHESIVES:



PVA-POLYVINYL ACETATE

This is a type of thermoplastic, which was discovered by Fritz Klatte in 1912. It is a white colour glue, which is suitable for glueing woods, papers and cardboards. It is strong after drying and remaining a transparent colour. It can be mixed with water so it could be dissolved again after dry. Some application of this glue includes bookbinding, wood binding, art projects and wallpapers glueing. Some of the PVA even is water resistance, which makes it much stronger. It is a non-toxic glue, which also doesn't have any smell.

SILICONE-BASED ADHESIVE

As glue contains silicon, a special structure of the compound, it is still elastic at a low temperature (-75°C) and high temperature (300°C) for short time. It doesn't normally react. They also resist water after dry. They cannot be used as paint since their surface tension is low. Normally they are used to bind plastic, metal and glass surface together. In the real-life, it is often used to bind parts in car engines or seal cables in electronic devices. It is chemically stable, which is not a toxic material.

HOT GLUE



A cost-efficient glue, which is very versatile. The hot melt adhesives are thermoplastic, which doesn't contain any water. When the glue gun melts them, they become to the liquid form, which then will cool down quickly and bond two sides together. It has a range of applications: packaging, craft, sticking labels. It also can last long and doesn't release any chemical product.



UNFORTUNATELY, ALL THE ADHESIVE ARE TOO DIFFICULT TO BE RECYCLED. I NEED TO USE LESS AND PUT ON THE LID ON TIME INCASE OF WASTING AFTER I USED THEM.

FURTHER RESEARCH

JOINTS:



very strong

PAINT:

Latex paint(acrylic paint):

pros: 1. Easy to apply 2. Dry quickly 3. Water-resistance after dry 4. Resistant to fading and chalking under the sun 5. Inexpensive 6. Not toxic

cons: 1. Needs to be sand and prime in order to apply them 2. **Need a large amount of energy to manufacture** 3. Difficult to recycle. 4. Releasing VOC

Chalk paint:

pros: 1. Eliminating the extra work(e.g. sanding or priming) 2. Dry quickly 3. Can work on any surface 4. Can be washed off with water and soap.

5. Environmental-friendly(consists of 30% renewable resources and 60% natural compounds.) 6. Not toxic 7. Low energy required to manufacture
cons: 1. **Needs to be covered with wax.** 2. Slightly longer for the drying process

Milk paint: (A beautiful powdered paint)

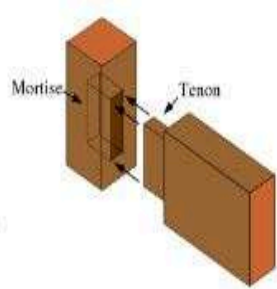
pros: 1. Able to sink in the wood(easy reuse) 2. Doesn't need pre-work for a finished product 3. Low price 4. Not toxic 5. Easy to recycle

con: 1. **Easy to chip off--needs a layer of wax to protect** 2. Short lifespan 3. Time-consuming

(This is the most strong joint in my opinion.

So I decided to research it further.)

MORTISE AND TENON JOINT



Use: sturdy frames, tables, doors, chairs, and other types of furniture

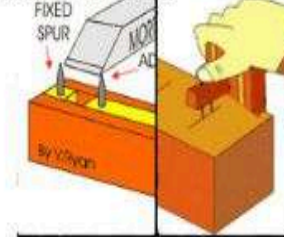
Sturdy rate: 95%

Requires: need to measure very precisely. Apply glue once put the tenon into the mortise and using the clamp to make it firm

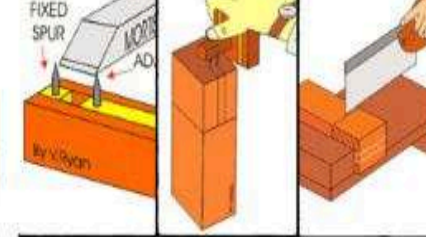
Making process:

1. Using a mortise gauge and a mortise chisel. Adjusting the spur on the gauge to make it match the width of the chisel.
2. Marking out the top and bottom of the mortise.
3. Using the mortise gauge to mark out along the wood.
4. Using the mortise chisel to remove the waste wood. Applying force with a mallet.
5. Doing step one again. Marking out the length of the tenon on the wood.
6. Using the mortise gauge to mark out the width of the tenon
7. Cutting off the waste wood with a tenon saw, with the support of a bench hook.
8. Chiseling out the rough edge of the tenon and then it can be fitted into the mortise.

MORTISE



TENON



Spray paint:

pros: 1. Dry super quickly 2. Easy to reuse 3. Easy to clean up afterwards

cons: 1. **A warm temperature is required** 2. little toxic 3. Releases VOC during the process of drying 4. Hard to recycle.

Oil-based paint:

pros: 1. **High durability** 2. **Glossy finishing.**
cons: 1. **Hard to clean (not water-based)** 2. **Needs to be sand and prime before painting** 3. **Needs to be seal with an oil-based finish** 4. **Toxic** 5. **Difficult to recycle** 6. **Requiring a large amount of energy to produce.**

(BECAUSE MOST PEOPLE LOVE THE PAINTED PLANT SHELF INSTEAD OF THE ORIGINAL COLOUR OF THE MATERIAL, I DECIDED TO INVESTIGATE DIFFERENT PAINTS.)

WOOD: PLYWOOD & flexible plywood

1. It is man-made wood, which decreasing the harms to nature
2. It is cost-effective.
3. Although it is stronger than other types of man-made boards.
4. The popular wood for both CNC and hand-making.
5. It has the texture of wood.



hinges: 90 degree hinge

1. It is easy to use and assemble
2. Other supporters are not required
3. Fit the size of the shelf.
4. Light in weight



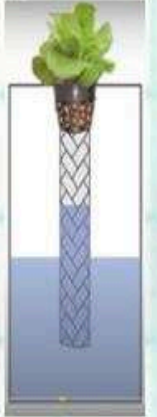
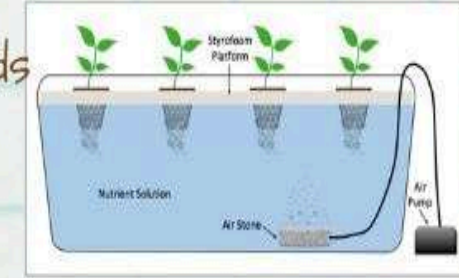
nails, adhesive and screws: choose during the process

I might use all types of them depends on my projects since they are all necessary

major machine & tools: laser cutter, 3D printer, files and band saw



hydroponic system: deep water culture & wick system



1. Both of them require less plastic reservoirs
2. They are much simple and easier than other types of system.
3. They require less essential equipment (e.g. water pumpers, gas pumpers.)

PLASTIC: PET & ACRYLIC & PLA

PET:

1. I love the process of upcycling. The most common plastic is PET so I will apply it to my design.

2. Recyclable
 3. Light in weight
- Acrylic:



1. transparent and strong
2. water resistance, which is not toxic
3. biodegradable
4. easy to work

PLA:

1. good filament for a 3D printer
2. biodegradable
3. strong outcome



major joint: mortise and tenon joint & dowel joint

1. Both of them are tougher than other joint
2. they look invisible once are applied
3. easy to work (CNC can help)

CONCLUSION

solar panel and plant's light: thin-film and LED grow light

1. They are cost-effective
2. They fit the normal plant shelf (size) and the hydroponic system (last longer, less heat)

Metal: low carbon steel

1. It is super malleable, which is easy to create curves or bend them.
2. It doesn't strong, but I will only use it as extra decoration since all the metals generally are not a suitable choice with water together.
3. It is a cost-efficient metal
4. eco-friendly

finishing: spray paint

1. a fast and effective way
2. less toxic substances are released.
3. cost-efficient



Design specification

Materials

Speak of material, I have to consider the properties of each component in hydroponic systems: The parts **contact with water should be water resistance**--not the wood indeed--but the outer frames could be metal or wood. The metal **must resistant to rusting or corrosion**. Plastic needs to follow my principle goal--protecting the environment: *'The less plastic I use, the better.'* I am an environmentalist and I have collected tones of plastic at home already for **upcycling**. Thus, I will apply them to this project (like net cups) to motivate people to take part in the upcycling program, which is also a cost-efficient way. In general, the materials should be **strong enough to hold water and other components, non-toxic and eco-friendly**.

Safety

The shelf itself **cannot be too heavy**, since it might cause any safety hazards. Each **corner of the shelf must not be very sharp**. The place with the reservoir should be **firmly attached** to the shelf, otherwise, it will spill out water.

Ergonomic

The shelf needs to have a **medium size**. It must be **convenient for people to reach the plants** on the shelf too. It should **stand firmly** on the table or the balcony in case of dropping.

Cost

The material and the system of the shelf are simple and sustainable, so the cost **should between 1000 Thai baht to 3000 Thai bath**. The overall price of the shelf is facing the mass market, which means **everyone has the chance to buy them** and work hard to dedicate themselves to part of the protection of nature. **Batch production is necessary**, which can decrease the waste of materials. The shelf itself is not fragile--shipping is not a big deal.

Functions

The product must have these as the **main functions**:

1. highly productive of food (more places for various vegetables to grow)
2. multi-functional (a drop leaf table/cabinets/boxes)
3. using renewable resources (solar panel)
4. able to grow in any area with different light intensity (LED grow light)

Optional functions:

1. have a place for a fish pod
2. have places for creepers to grow
3. can check the water level easily



Aesthetic

The product needs to be related to the form of nature in order to promote a sense of protecting nature. The overall shape of the shelf needs to show a **mixture of nature and modern design**. Its **colour will vary**, mainly blue, green, yellow or the simple black and white. However, it has to **change for different purpose and customers**. For example, if people from a low-income family, trying to grow food effectively, the shape of the shelf don't have to be too fancy.

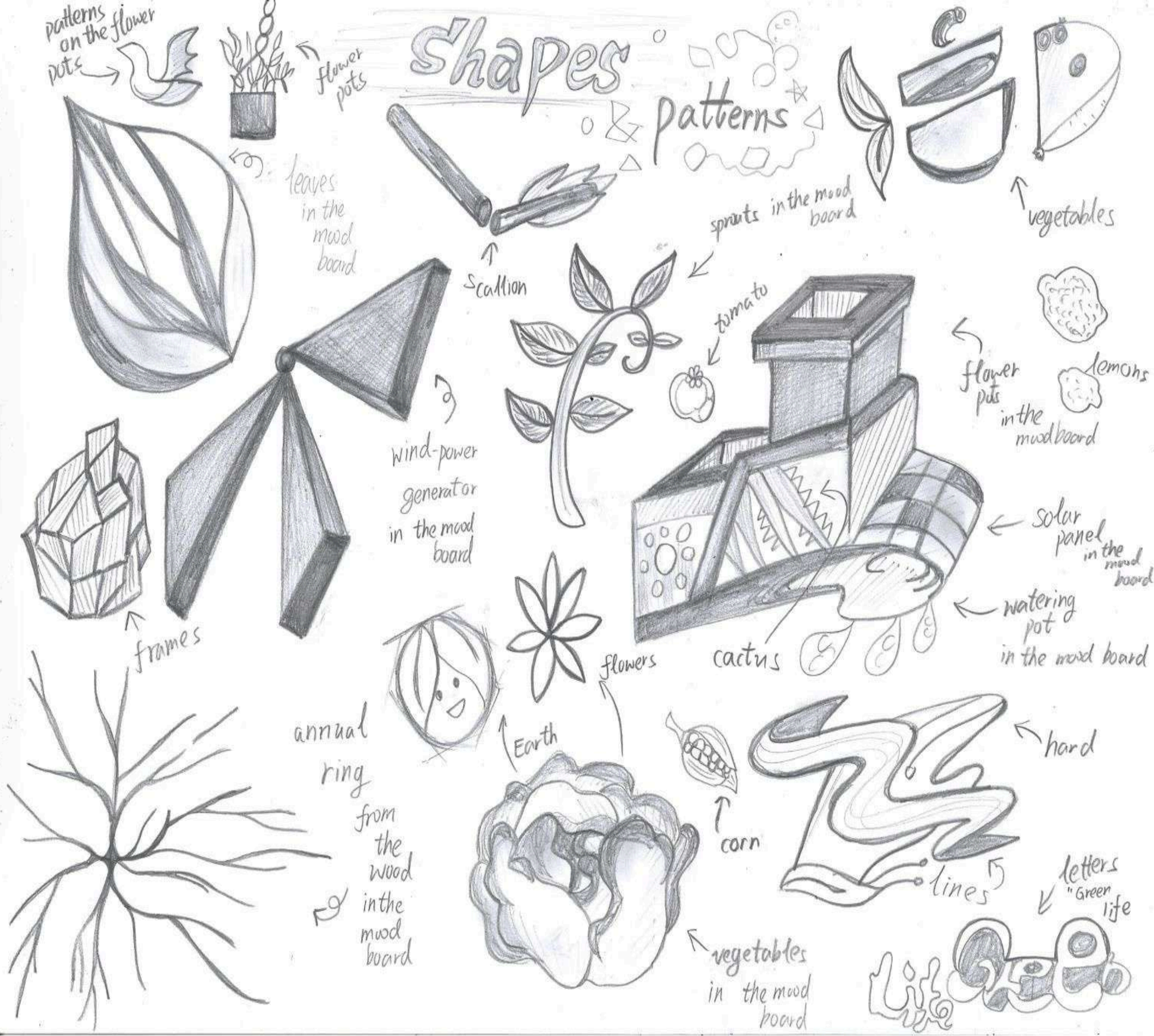
Manufacture techniques

It must include various techniques, mainly use CNC since the size of the shelf should be precise. The components need to be **joined in a simple way**, which also can be constructed and deconstructed easily. On the other hand, I need to add a **solar panel and a LED plant light** on the shelf. I want to decrease the chance of wasting non-renewable energy. The light will also allow plants to **grow in the dark**. **3D printing** is also a great technique, producing water-resistant containers.

Client

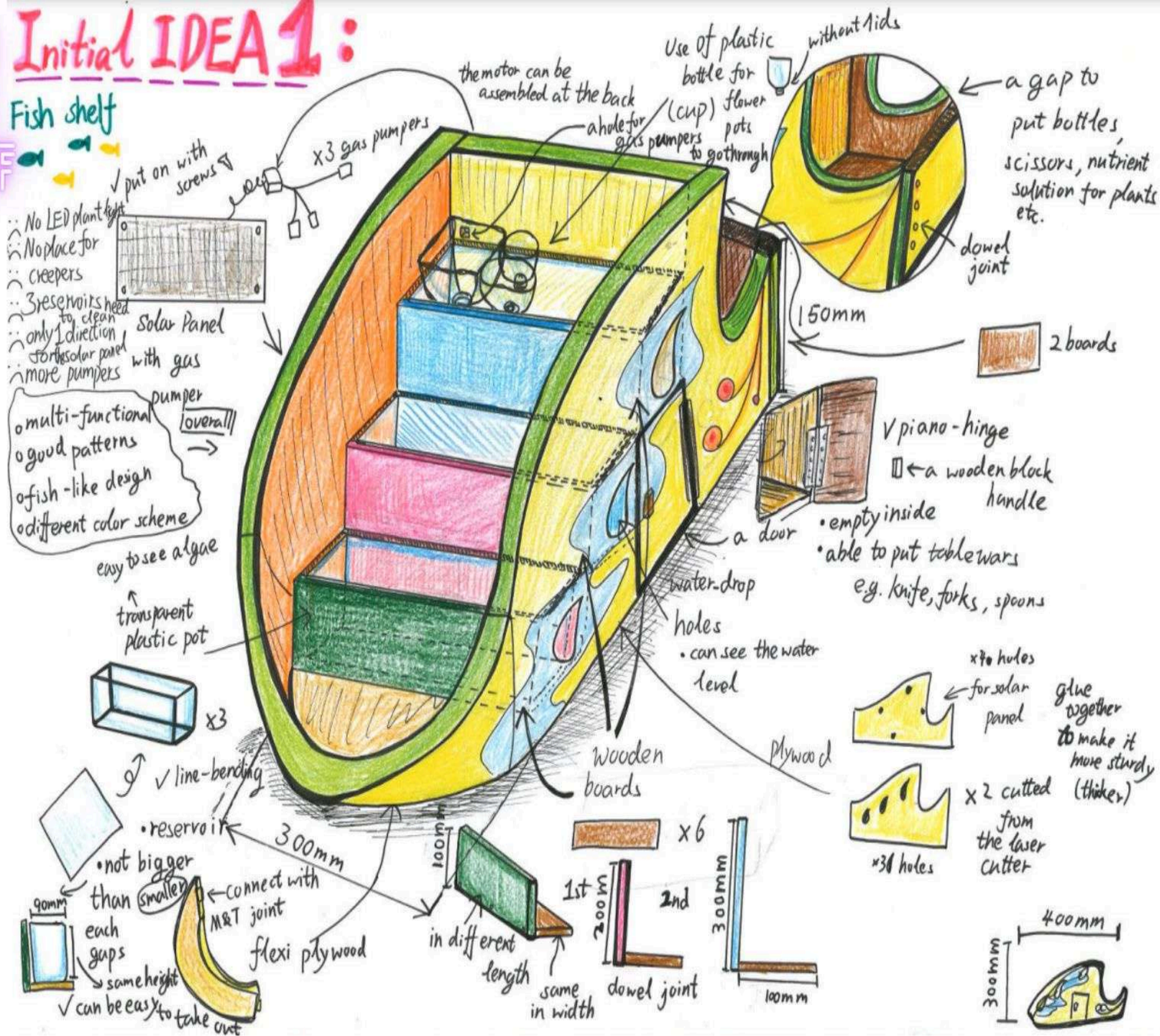
This shelf faces the **mass market**, which is mainly the **adult citizen** or the family who want to have their own gardens. However, they should **have a basic knowledge** of the hydroponic, which is also an option for people who are willing to learn gardening and new technology.

SHAPES



Initial IDEA 1:

Fish shelf



Initial IDEA 2:

little house ↑

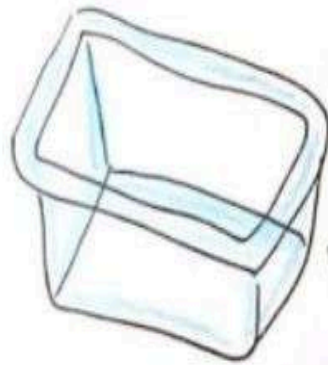
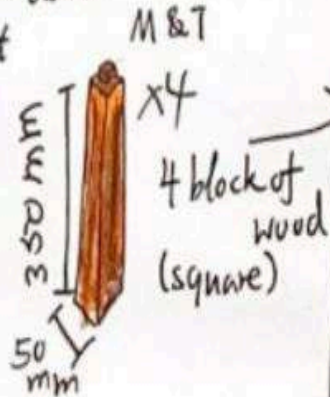
HOUSE LIKE SHELF

It follow a generally mature design with wooden colour.

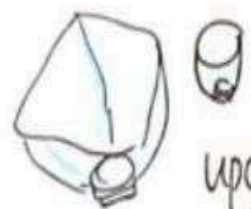
- ~ No place for creepers
- ~ more reservoirs need to clean
- ~ more pumpers needed
- ✓ different colors, patterns

- ✓ allows plants to grow in a dark environment

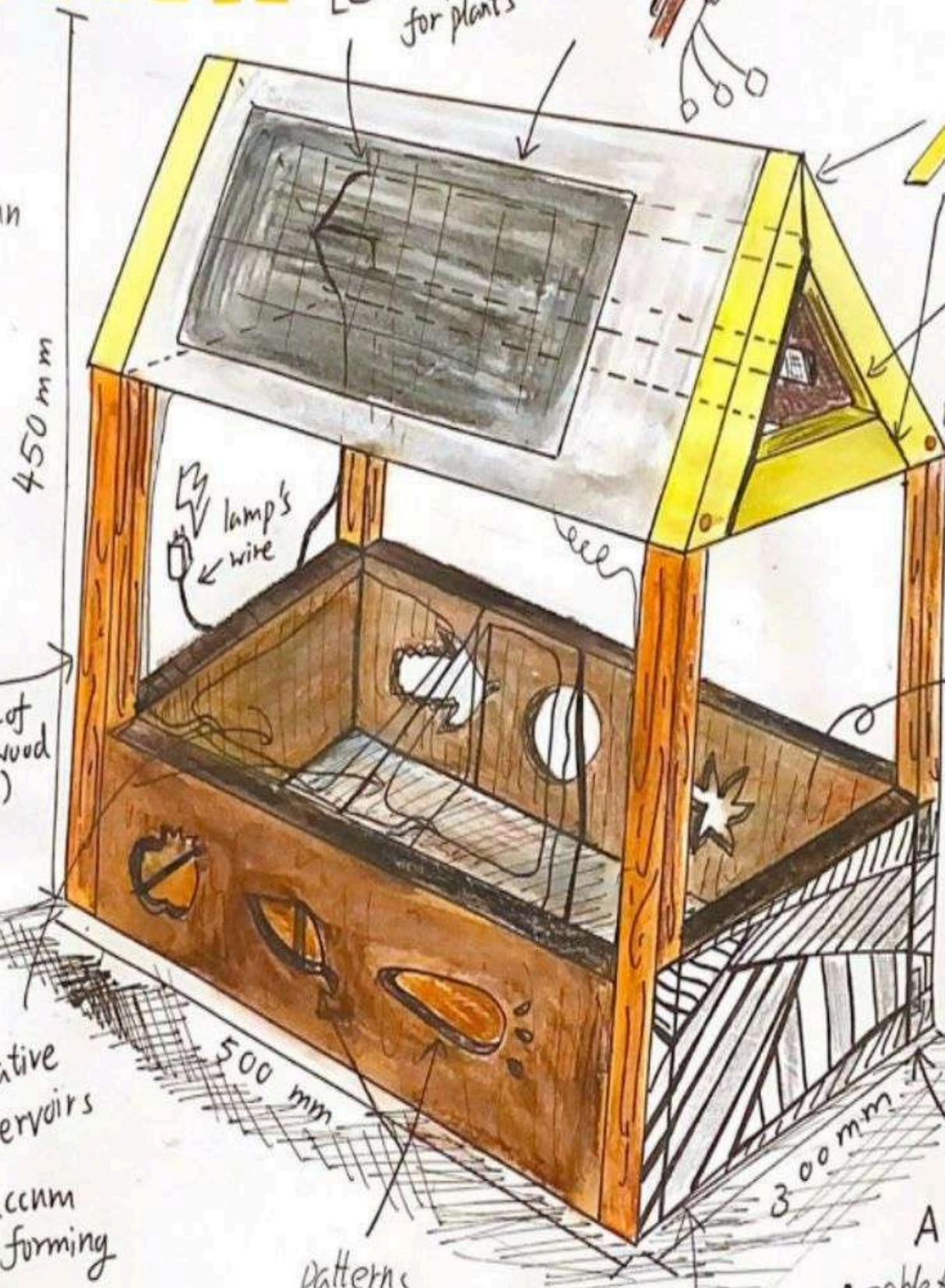
✓ house-like design



✓ vacuum forming



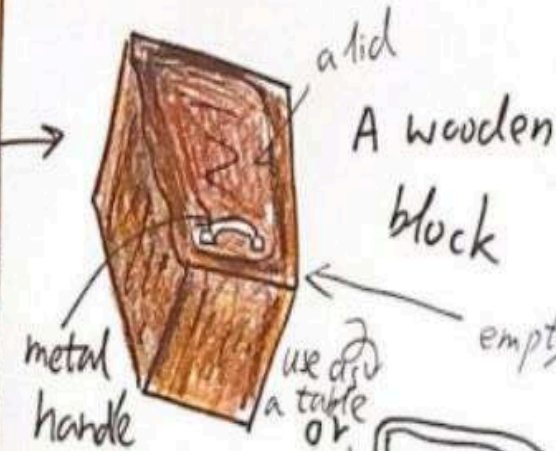
✓ transparent



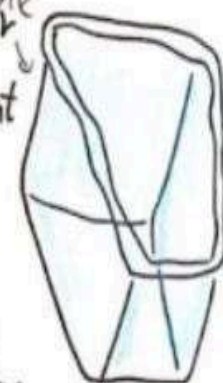
Miter joint

the hole

dowel joint
Mortise & Tenon joint



• a transparent reservoir



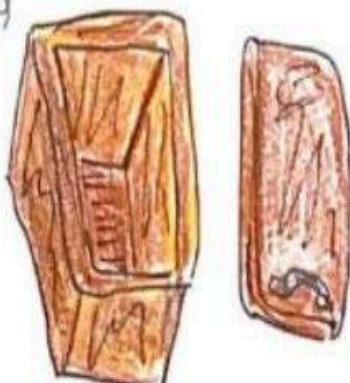
• wooden box

A door

• -able to open and get things out
• able to put things inside

Bottom (butt joint)

patterns as holes
• can see the water level
laser cutter



out of

• put anything, also plants

Initial IDEA 3:

Call of nature

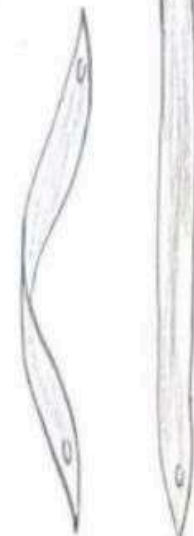

BIRDLIKE SHELF

This design combined more shapes and colours of nature, which also have the good racks for creepers to grow.

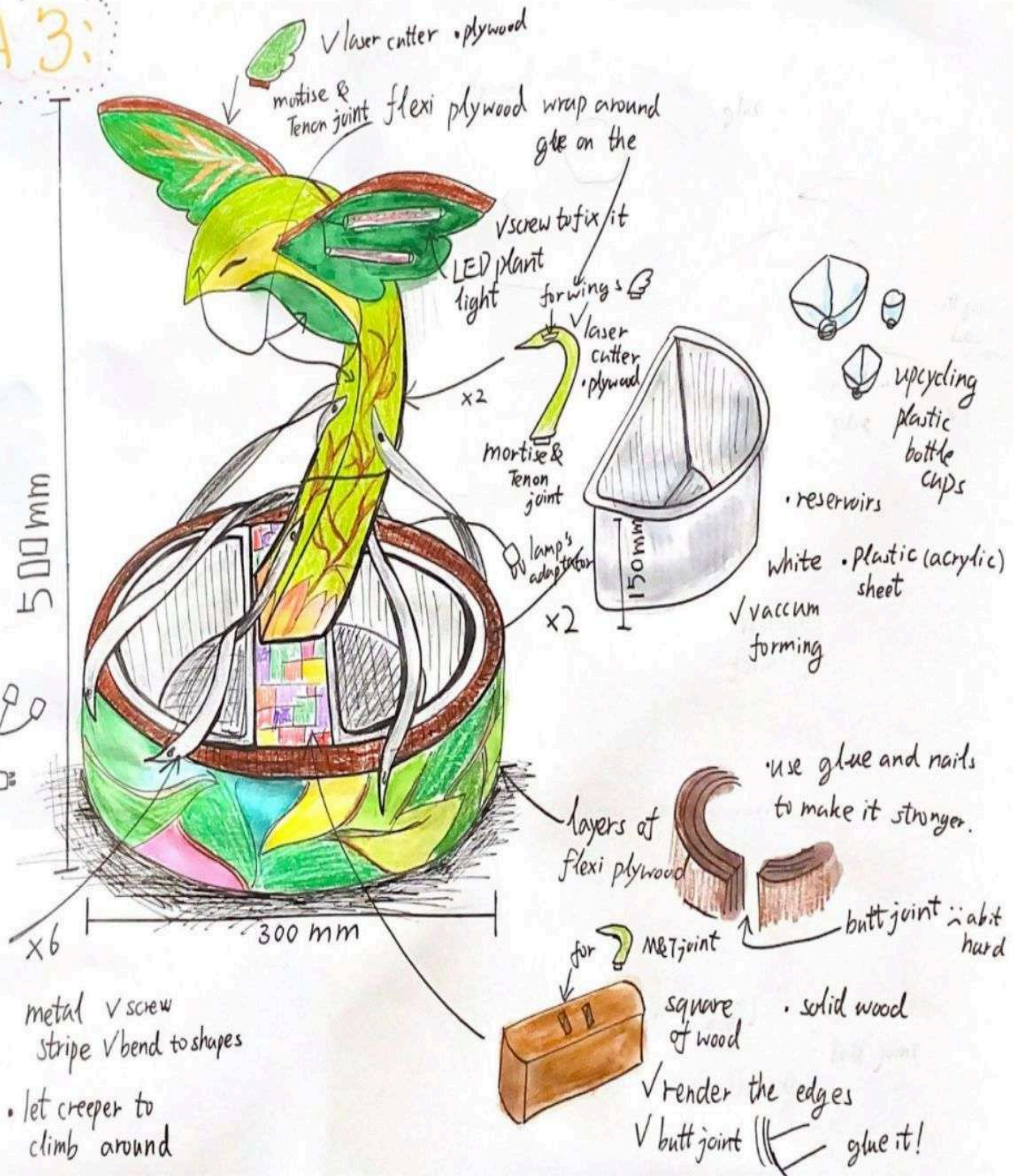
- ✓ great patterns
- ✓ High ornamental value
- ✓ can grow creepers

⚠ You need an extra electrical gas pumper since the solar panel doesn't fit.
 ⚠ hard to construct

• Low Carbon steel



metal V screw
 stripe V bend to shapes
 • let creeper to climb around



Initial IDEA 4:

Blue Owl

It is super multi-functional, with the combination of all the features, including shapes and colour, functions and shapes.

- A bit big
- creepers can grow
- have places for solar panel and LED plant light

Summary:

- lots of small boards join together
- V shape is novelty
- multi-functional flower pots

upcycling flower pots

mortise and tenon joints only

flexible plywood make curves

A hole which can help user to see the water level

places for putting chopping boards, a knife (knives), scissors, seeds forks, books (menus)... etc.



Need a hole for wires to go through

Maybe I can use Mortise & Tenon joint for this.

+ glue corners

✓ line-bending

✓ transparent acrylic board

LED plant light + screws

on a board since the top is shorter than lamps

motor & pump

laser cutting 2 big boards x 2

put long wire inside

make a hole in this table

90° lift-up / drop table

✓ solar panel

✓ 90° hinges

can change angle to let the solar panel face the sun

glue a sheet of wood to hide the joint lines

✓ plastic pots for a small creeper

wick system

fit a fish pot

no need for nutrient solution

✓ glass

→ a quaponic if you change water

✓ steel (LCS) handles for creepers



Interview

I discussed my initial ideas and showed them my specification. They added more ideas to them. I was really happy about that.

possible client 1

First I interviewed my Biology teacher, who love vegetables so much. After I showed her my ideas, she was pleased with them." *I quite like idea 1 and idea 4.*" she said, "I think your shelf always need a **place to put scissors for cutting vegetables.**" Based on her advice, I believe my final design needs to include such a function.

Moreover, we discussed the colour scheme."I like **light colours** and they can make me happy."I personally love the lighter colour too. In the end, we talked about production and size. All my ideas are suitable to put on a table, but none of them can provide enough food for a family of more than 3 people. I agreed. Since it is a desk-top shelf, we all don't expect it can provide a large amount of food. In the end, she asked me to **think about which one is possible to construct**, which is more important than functions and colours

After interviewing my Biology teacher, I went to talk to my computing teacher, who actually like gardening. "Interesting ideas! I like your design." he agreed with all my ideas but chose *idea 4* as his favourite one. I asked him the reason. He told me the **combination of**



possible client 2

hydroponic and aquaponic is smart. With the combination of LED light and solar panel, it makes the design stands out. I believe he means technology is a great thing in this type of shelf. He asked if I will put a program in this system, making it become automatic. Unfortunately, I couldn't add any programs to it because I study resistance materials and knowing less about constructing a system. He understood me and also give me some construction ideas.

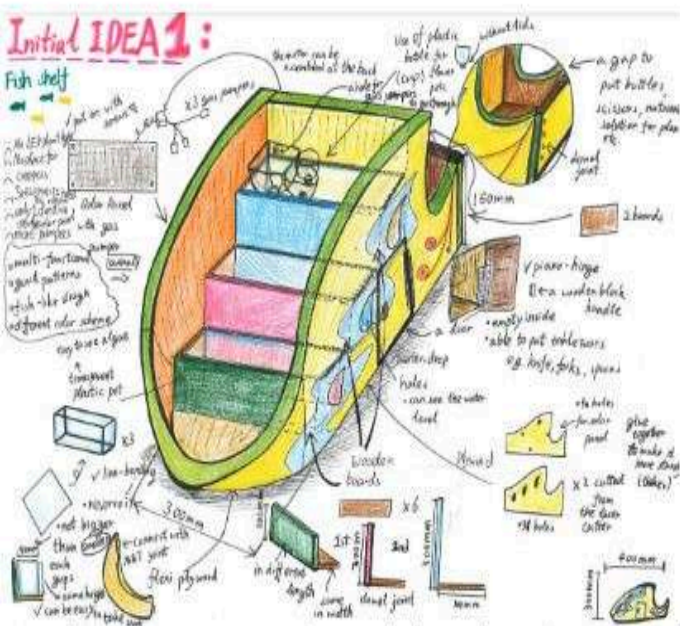
At last, I visited a local flower shop, asking the seller for some properties of vegetables, fruits and flowers. That Ms explain everything kindly to me, even including how long to water them. She said most of the vegetables **need to have some poles to support and climb on.** Not only creepers but also normal vegetables such as tomatoes and cucumbers. I soon showed my ideas to her.

She like idea2 and idea 4." If I will buy one of them, I will choose these two ideas. I love flowers and most of them need spaces for climbing too." After the discussion, I get a better idea about what I must include in my shelf--A place where the creeper can growing around.

possible client 3



IDEA 1 RATE:6/10



This is an adorable design with light and relaxing colours, like stairs. (hydroponic only)

IDEA 3 RATE:8/10



It looks incredibly cool to me, give me a sense of freedom(hydroponic only)

FINAL DECISION

Wood and plastic fit each other very well but there is no place to apply metal.

The sharp edges are smooth and the materials are not toxic. The reservoirs are firm inside.

It has 3 separate areas for plants to grow, which is easy for people to reach.

It uses many techniques, with the bending of plywood and paints, which should around 1000-2000 Thai baht.

There are two spaces for putting utensils: the box and the cabinet. There is no space to put grow light.

It looks perfect in shape and design with natural patterns and fish shape.

Honestly, the joint and flexible plywood is hard to apply, including the area for a solar panel.

Younger clients. It cannot produce too much food overall, which doesn't fit to a huge family.

All the materials are perfect and fit the size of the shelf. The metal has no place to use as well. It is sturdy but the pointy edges are presented.

Water is not easy leaking.

It is a perfect size with great space for plants to grow, although the box might a bit hard to put in.

The LED light and the solar panel for the gas pumper are all applied, so the price isn't low.

It doesn't have too much space for other functions, except the wooden block with a container inside, which can be either taken out or use as an empty container.

The shelf is simple and attractive, with vegetable shape holes on the bottom.

It uses dowel joints or butt joints are more often=needs to be cut and assemble with great cares otherwise, the bottom might drop because of weight. This shelf fits for a family with medium amount of production.

All the materials are used in this design, including the 90-degree hinges for a table.

A super strong design with some sharp angles, but everything is fitted firmly, including the place for reservoir. It might be too tall to put on a dining table, but it is great on a coffee table. There are two spaces for plants to grow.

It contains lots of components and many other functions, which is expensive overall(>2500THB)

It contains the metal strips for plants to climb at the back, a place for fish pod and a place for putting cooking wares. The LED light and solar panels are used.

It is simple in colour: blue can match with the green plants very well. The shape is interesting too.

The mortise and tenon joints are the major joint in this project, which is quite challenging

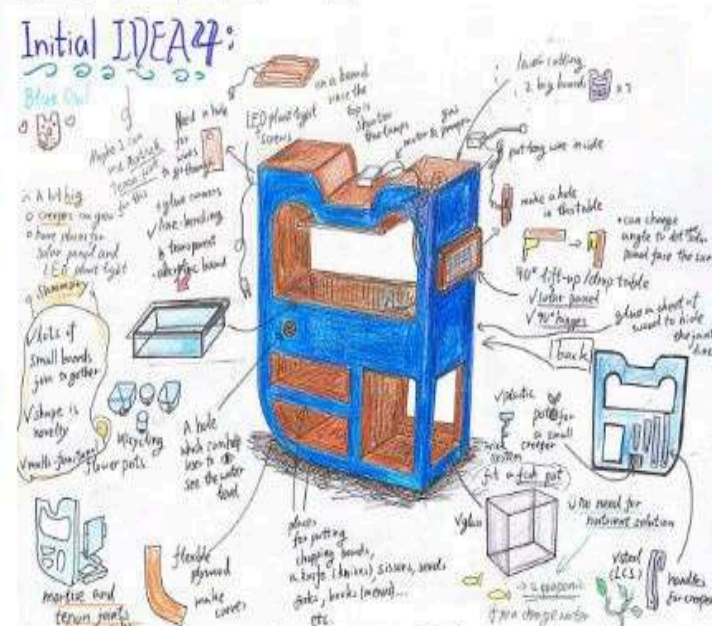
This is not the most productive shelf, but its functions could attract more client, from young to old

IDEA 2 RATE:7/10



The simple design with all the features--a feeling of home:) (hydroponic only)

IDEA 4 RATE:9/10



I love the shape of the shelf so much. The owl is my favourite animal after all.(hydroponic and aquaponic)

paper model

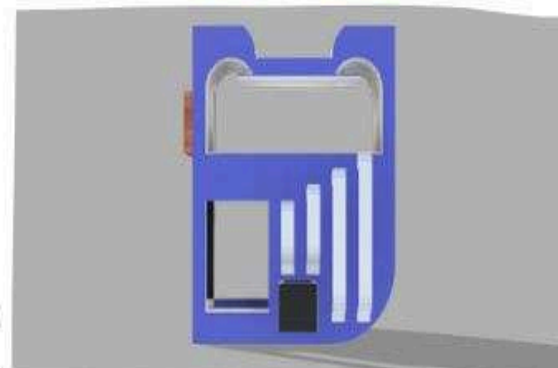
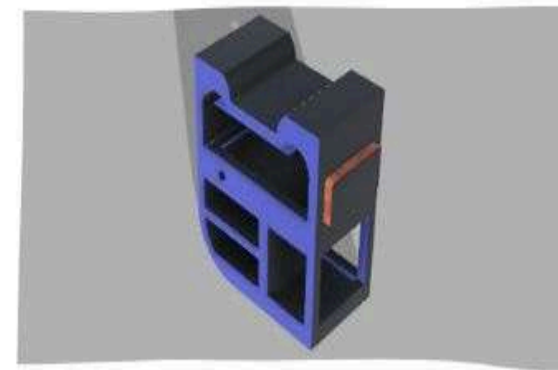
I choose initial idea 4 based on the comparison of other ideas.



I create a card model for the idea. Although it doesn't look perfect (tapes + card papers), it still makes me consider a lot about how to manufacture this special shelf. During the modelling, I cut off two main front shapes of the shelf and glue them with papers in between, which can be applied as a construction option. Because the flexible plywood is quite thin, so it needs to glue on a flat surface of thick wood to perform a bend. Currently, this model's side is too thin to glue the outer frame of flexible plywood (paper). After checking the size of this 1:10 model, it is a bit too high for a desk-top shelf, which means it can be slightly difficult for young people to reach them. For the back, I create thin pieces of paper (metal strips) and a box, which give me an idea of the curves' shapes.

I also check the size of cabinets, including the area to put fish out. They all should fit their function. The top curve is a bit tricky to make, but I manage to cut a small piece and stick it on. In the real making process, I might do the same.

CAD MODEL



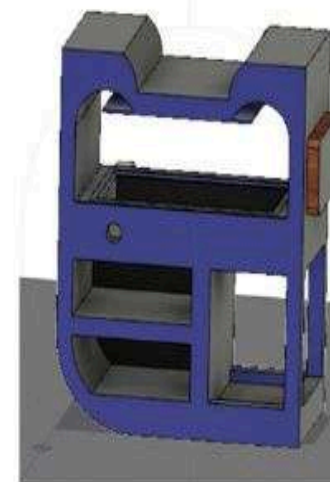
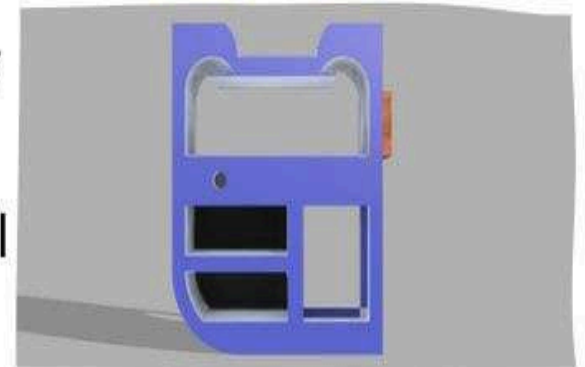
Thinking of the solar panel, metal strips and LED grow light, I know that I have to mark out the place where the screws go for this design and provide enough space to assemble them. From making this product, I learn basic modelling techniques.



After I change the shelf's height, the 1:10 CAM model looks so much better than the paper one. Back to the manufacture, I must make sure that each part of the components is sturdy and fit my desire shape.

Except for the front board, I still need multiple small boards including sides, tops, and bottoms.

Considering joints, I believe the mortise and tenon need to be cut precisely with the laser cutter later in the process. The model shows the outcome of my project, which provides me with rough ideas of size and shapes.



TESTINGS

test 1: laser cutter and strip heater

Based on my research on machine and materials, the acrylic board is the one that I chose to use. I would like to use the strip heating techniques because the vacuum former has the limitation of the size. At the same time, I learnt how to use the laser cutter, which is convenient and accurate. I am a bit of worry about the permeability of the heat-bending plastic reservoir since the glue might not perform super well.

Firstly, I use RD Work, a CAD software, to draw an out frame of a small box. It is difficult to use for the first time but once I grab the techniques, everything becomes easier. After drawing the outer frame, it is important to make sure all the lines are connected to one another to form a shape. The edges can be chamfered to make it easy to bend later. When I press the start, the laser began to work, cutting the acrylic board smoothly. It is important to stay around the machine in case it might catch the fire. The cases of the laser cutter need to be close at this point to protect the eyes. When the "Bip, bip" sound appears, it means the machine finish the cutting works. Before taking out the entire thing, we need to double-check if the piece is cut off successfully. Sometimes the laser cannot cut through the entire material. If it is be cut successfully, the brown paper on the bottom of the board needs to be removed. Preheating the strip heater and waiting for it reaches above 375-degree celsius. Then I place the places on the board where I want to bend. As the temperature increase, the board becomes flexible and hot. Insulating gloves are required. Bend the edge with two pieces of wood and move the board away from the machine, it will begin to cool and become stiff again. After all the corners are bent, I use silicon-based glue to glue each edge together. When I test it after the glue is dry, it is quite good with great water resistance.

test 2: possible upcycling materials

As I mentioned before, upcycling is a wonderful recycling technique. I decided to test different types of plastic bottle top to use them to make the flower pots for hydroponic. In my area, we must drink bottled water because the tap water contains too many chemical compounds. At first, I don't really care about those bottles but because the education about the environment is increased in social media and school, I began to focus on it. In the early of 2020, followed by the lock-down, I began my 'upcycling' life: Chairs, shoe shelves, fish pots, even bed. It is fun in the process.

Also, the properties of the plastic bottle are useful. They are light and water resistance, which are non-toxic as well. I try to identify their length and diameter in order to design a specific size of reservoir for my project. It needs to cut along the top of the bottle and measured by rulers. The way to assemble each pot together is stapled or tightened with ropes.



test 3: the flexibility of flexible plywood

As another major material in my project, flexible plywood can only bend in one way. It is indeed flexible but all the wood has a limitation, including it. I use the protractor to measure how much it can be bend over. It turns out that if I don't move the bottom part at all, the wood only can reach 20-degrees upon each side. If only one bottom side is not moved, the wood can be bend more than 100 degrees. In conclusion, it is really flexible wood.

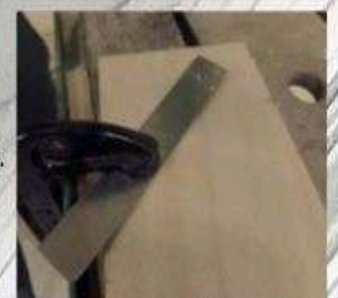
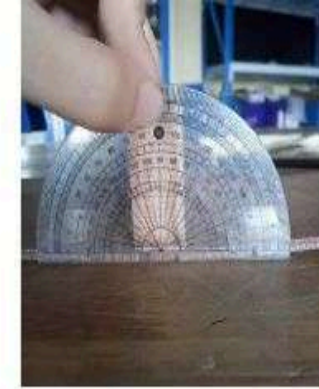


However, I investigate what do people do if they don't have flexible plywood in the first place. One alternative way is glueing small pieces of wood together along a curve, which shows on the right side. I don't think it is an effective way but it is tougher than just a a layer of plywood. So I decided to combine tow together to create a strong curved surface.



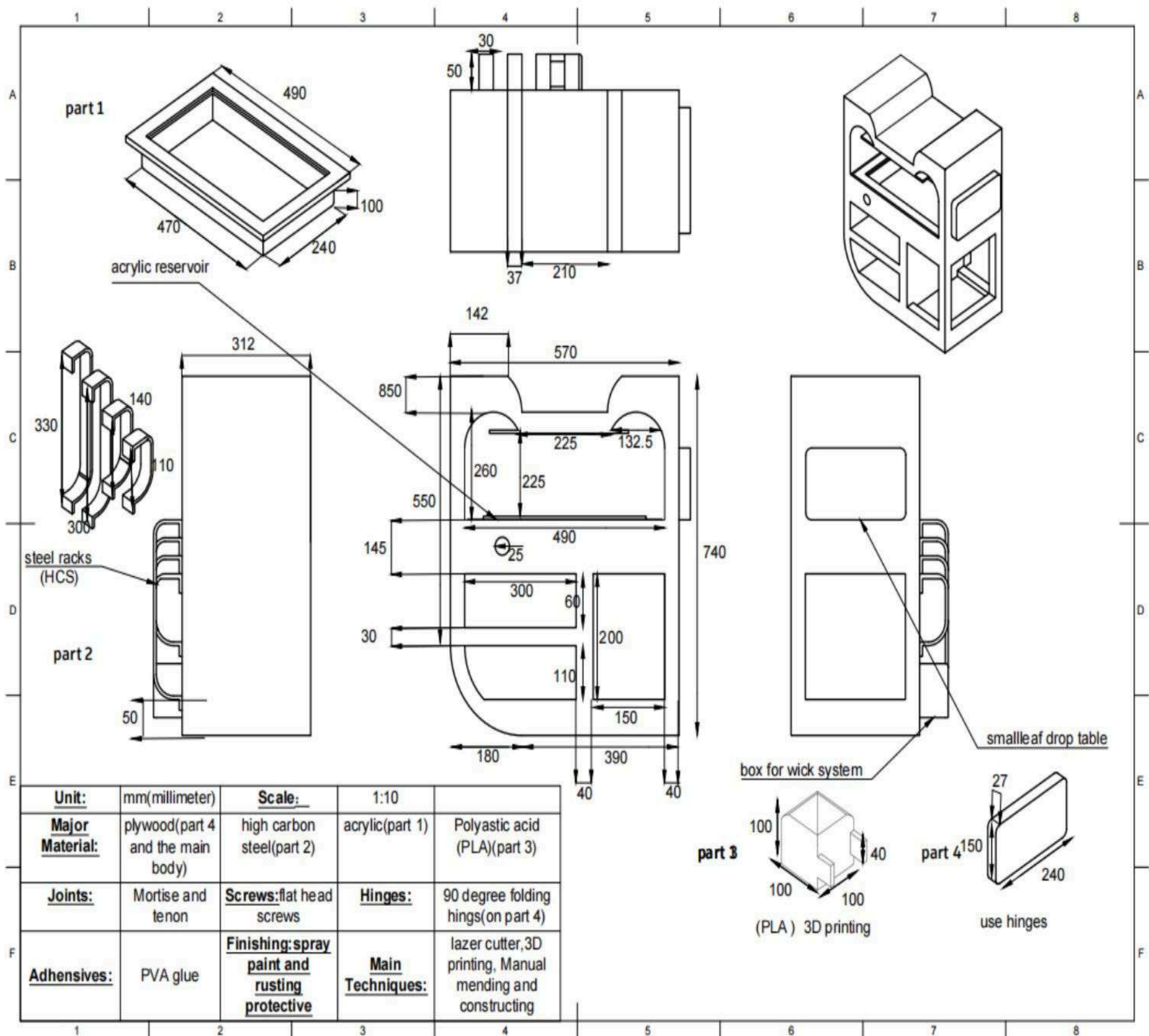
test 4: pillar drill and center punch

The metal strip is another material that I will testing. Because I will assemble them as a rack for creepers, the way of assembling is an important thing to check and test. After a metal sheet is cut into many strips, they have got super sharp edges. They need to be smooth before use, (I don't want my client's fingers to be cut off) I am only testing how to put a screw perfectly in it. Followed by technicians' instructions, I use the centre punch and a mallet, give the metal surface a strong hit. A concave appears on the metal. The purpose of it is to make a concave for pillar drills to hold it. Then it is time to drill!. When we using the pillar drill, we cannot put hands on the working pieces because it could spin as the drill bit goes through. A clamp and a scrape of wood are required, which is used to hold the materials and prevent the driller from drilling all the way through the platform

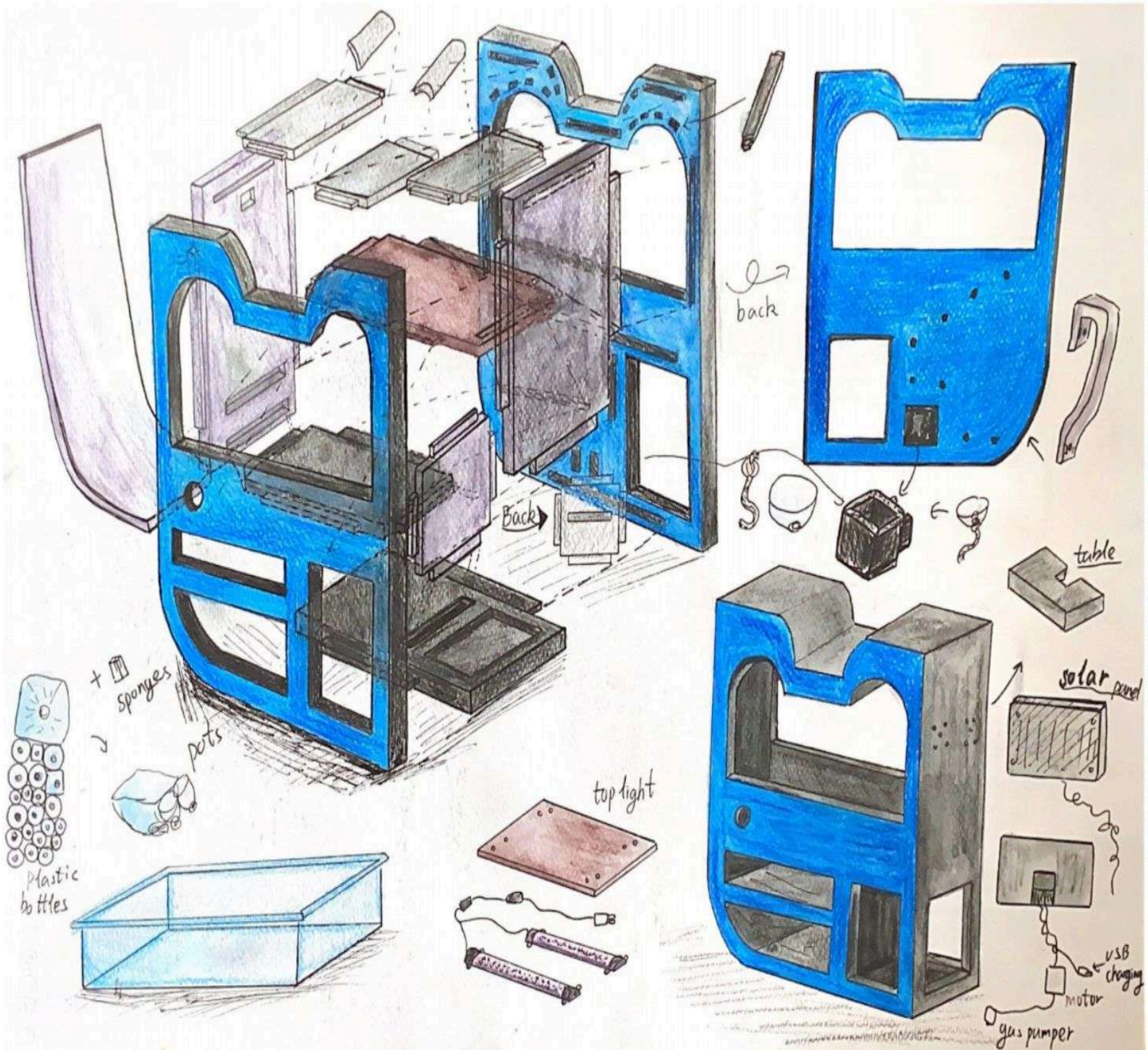


Process	Equipment	Materials	Health and safety	Quality control	Time
<u>Preparing all the major components</u>	<i>Laser cutter, (wood:) PVA glue, wood files, 6 clamps (plastic:) strip heater, silicon based glue, masking tapes, heat-resistant gloves</i>	<i>plywood</i>	<i>Wearing heat-resistant gloves while bending</i>	<i>Measure the exact 90 degree angle while glueing all the pieces together</i>	<i>4 weeks</i>
<u>3D printing the back box for wick system</u>	<i>3D printer, masking tapes</i>	<i>PLA</i>	<i>Make sure the machine is unplugged after using</i>	<i>Make sure the plastic filaments are not stick together to create inaccurate shapes</i>	<i>3 days</i>
<u>Prepare metal strips</u>	<i>Circular saw, metal guillotine, metal files, abrasive papers, masking tapes, pillar drill</i>	<i>Low carbon steel</i>	<i>Pay attention to the sharp edges of the metal strips</i>	<i>Check the edges of each metal strips are smooth and the corner are blunt</i>	<i>4 days</i>
<u>Prepare flexible plywood</u>	<i>Table saw, band saw, file, sanding machine, goggles</i>	<i>Flexible plywood</i>	<i>Wearing goggles while sanding</i>	<i>Check each sides of the pieces are accurate</i>	<i>3 days</i>
<u>Creating upcycling plastic pots</u>	<i>Stapler, craft knife.</i>	<i>PET bottles</i>	<i>Careful while cutting</i>	<i>Checking all the pots are at the same height</i>	<i>4 days</i>
<u>Painting/coating all the pieces</u>	<i>Spray paint, mask, metal primer</i>	<i>Spray paint</i>	<i>Wearing masks</i>	<i>Make sure everything are be painted/coated</i>	<i>3 days</i>
<u>Assembling all the metal strips, solar panel, plastic pot and grow light to the wood board</u>	<i>Screwdriver, scissor</i>	<i>Flathead screws, spongy, washers, solar panel, grow light, 90-degree hinges</i>	<i>Careful while drilling and cutting</i>	<i>Keep all the spongy is the same size and all the screws are carefully assembled</i>	<i>5 days</i>
<u>Assembling all the wood pieces and flexible plywood together</u>	<i>Pin hammer, PVA glue, chisel, sanding machine</i>	<i>Box nails</i>	<i>Careful while hitting the nails in</i>	<i>Making sure they are all fit to each other</i>	<i>6 days</i>





PICTORIAL DRAWING

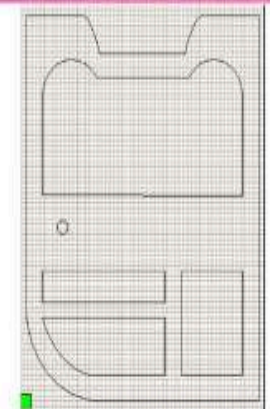
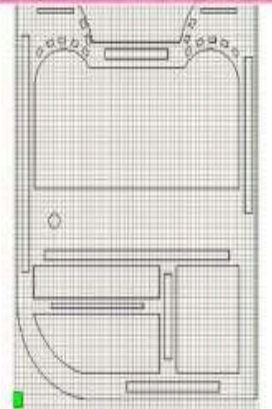
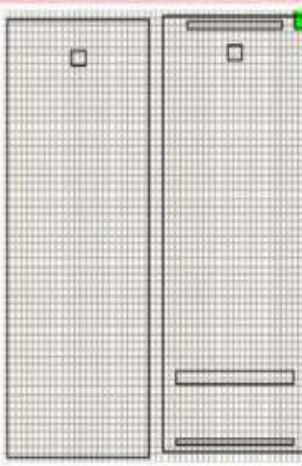
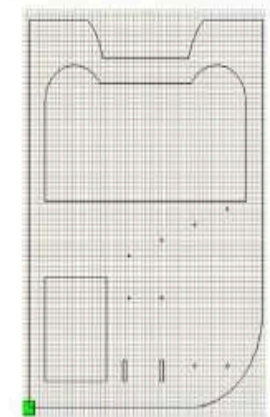
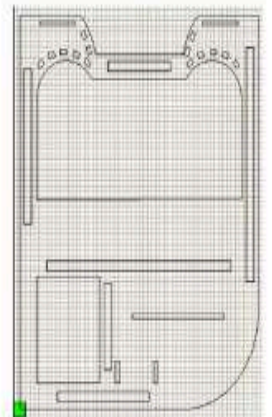
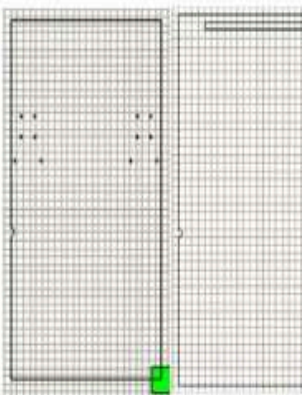
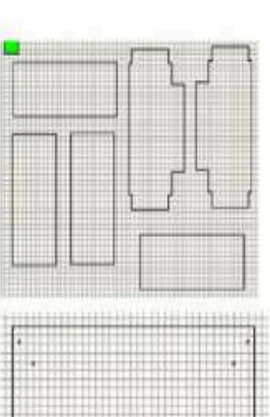
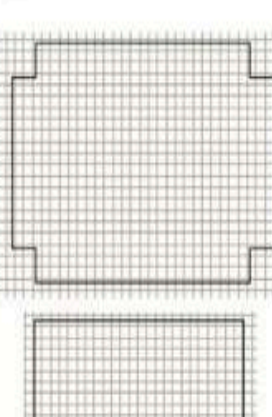
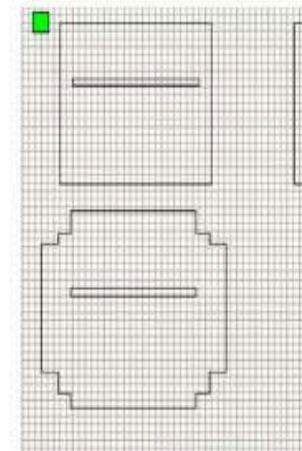
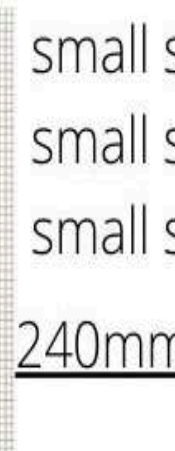
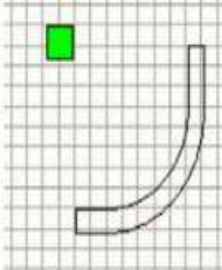
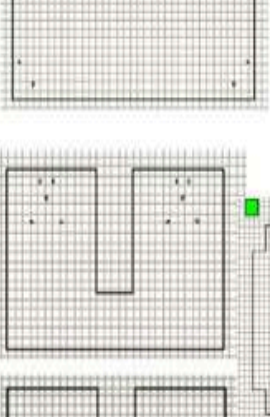
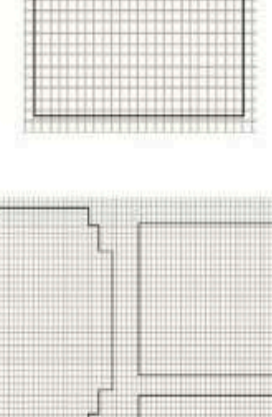
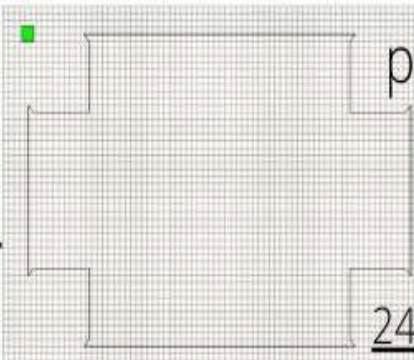


cutting list

after separating each part of the shelf, I created a clean scale for each component. 9mm is the thickness of the wood that the laser cutter can cut through, which is too weak for a shelf. Since I want the mortise and tenon joint to be a success, I have to glue each layer of the wood together to make it much thicker.

brainstorm of
my ideas

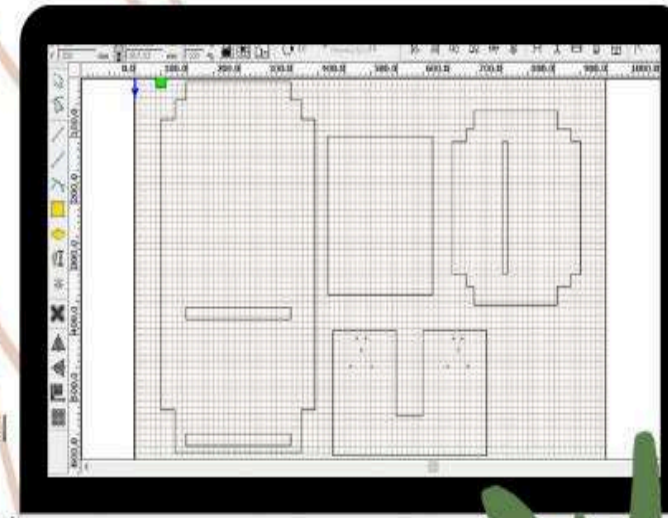


		front board 1 x1		left 1 x1		left 2 x1		left 3 x2		strips x14
		front board2 x3		left 2 x1		left 3 x2		right 1 x1		the curve for metal bending
		back board1 x1		right 1 x1		right 2 x1		right 3 x2		x4
		back board2 x3		middle 1 x1		middle 2 x1		middle 3 x2		plastic reservoir x1
		<u>570mmx740mmx36mm</u>		<u>240mmx550mmx36mm</u>		<u>240mmx450mmx36mm</u>		<u>240mmx570mmx36mm</u>		<u>240mmx30mmx9mm</u>
		up small boards 1 x3								<u>120mmx36mmx9mm</u>
		up small boards 2 x3								
		top board1 x1								
		top board2 x2								
		top lamp board x1								
		<u>240mmx142mmx27mm</u>								
		<u>240mmx225mmx27mm</u>								
		<u>240mmx360mmx9mm</u>								
		table with joint holes x1								
		table without joint holes x2								
		middle cabinet1 x2								
		middle cabinet2 x1								
		<u>240mmx150mmx27mm</u>								
		<u>240mmx300mmx27mm</u>								
		small supporter1 x1								
		small supporter2 x1								
		small supporter3 x2								
		<u>240mmx200mmx36mm</u>								
		base 1 x2								
		base 2 x2								
		base 3 x1								
		<u>240mmx390mmx45mm</u>								

LASER CUTTING



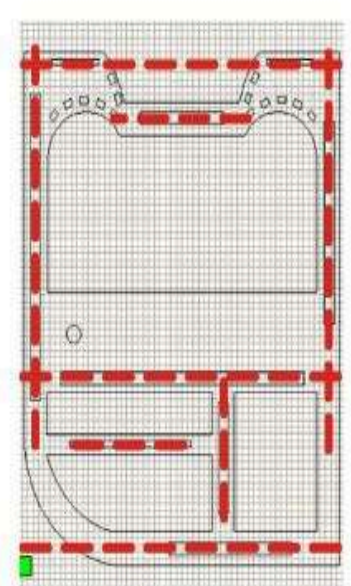
To begin with, I use RD Work to create all the wooden pieces that I want. I use fine straight lines to build all the shapes for mortise and tenon. They need to be measured incredibly accurately, including each of their positions. on the board.



Once the glue is dry, the wooden components' shape and angles cannot change. If there are any mistakes in the joint, they need to be recut or get rid of the parts that are not necessary with chisels, which is not a good choice. Luckily, every single component is correct in size and shape, making them easy to assemble. This cutting and glueing process almost took me 3 weeks, which is the most crucial step in the whole making project. Clamps are easy to run out when the pieces are larger, especially when I am glueing the front and back pieces---I used all the possible clamps in the classroom! Followed by every single wooden components' manufacture, The laser cutting, step 1, is close to the end.



The front board is the one that is hard to create. It contains more than 15 mortises and all of them need to be positioned in a reasonable place. Unlike other components that I created later, the front board is the most essential part of the whole project. In order to make each joint strong and powerful, I make the mortises in the middle of the jointed parts.



In the stage of cutting, the size of the wood needs to be as same as the size of the laser cutter's working area. Plywood will be cut if it doesn't fit the size of the machine. After it is placed in, the laser's distance should be modified as well depending on the thickness of the material. During the cutting process, I must stay around the machine in case it caught fire.



When all the pieces of one component have been cut off, it is time to glue them all together. I use PVA glue, which is the best choice of glueing wood surface together in layers. A thin layer of glue will perform very well. It is important that the glue can be squeezed out and block the mortises, which means I must put less glue on the areas around the joints. After all the layers are put together, I need to check if each layer is completely fit one another.



After checking every single layer, which is fitted together seamlessly, it is time to clamp them. Because each corner and tenons need to be glued together firmly, different types of clamping tools are used: G clamps, quick-remove clamps and spring clamps. I put other wooden pieces under the clamps' heads, preventing the clamping areas to get some patterns and distributing the force evenly around the areas.



Eventually, the wooden components are changed and glued, which are used as separate joint boards. They will be used as an adhesive board to one another. Once all the boards are cut off and glued, it is time to test if each joint are fit on each other.



CURVE FOR METAL BENDING





After cutting the wooden components, I move on to creating the plastic reservoir with strip heating. When the transparent plastic acrylic board is placed into the laser cutter, the cutting process started. I cut it multiple times to get the desired shape entirely off. I removed the protective paper that the board has before the cutting because it is easier for the laser to cut through. The shape I designed is a regular box with extra edges on each side. It is used for getting out the reservoir and prevents water split into the wood since it covered some area of the wooden frame.

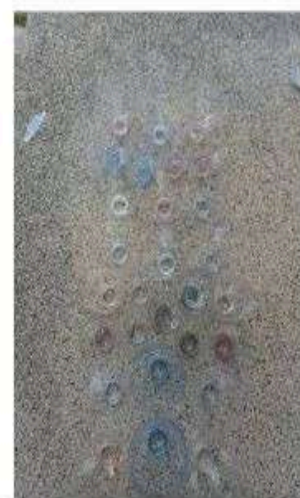
I preheated the strip heater, increasing the voltage so the temperature can increase faster than before. At the same time, I measure the side and the length where I will bend on the board for later use. By sticking some paper masking tapes, I get to know how much and where to bend it. When the strip heater reaches 380 degree Celsius, I put it on with thermal resistant gloves. No more than one minute, the plastic board becomes elastic that I can even bend with my finger.



Thus, it is the optimum time for bending acrylic. I hold one side tightly, placing a long piece of wood underneath the small triangular edge, lifting up the entire board and pressing quickly with all my strength in a place that is far from the



Upcycling of plastic plat pot sounds a bit wired to you? It is not weird at all. Don't worried about its lifespan, PET bottle is more flexible than you think. I choose them because I am an "old friend" with them. After cutting the top parts of different sizes of the bottle off, I poke multiple holes on it as the holes form the net cut for the real hydroponic system.



I try to use hot glue to glue each cups together at first, but that option doesn't work because the PET will melt and change the shape in the hot temperature. So at last I choose to use staplers to staple them. It ends up with a good result. The cups look neat and the shape is still doesn't change. However, it might not as strong as being glued. The hole in the middle of the cup is used to put cotton for hydroponic, which can contact with water the most. For the wick system, the system with a rope that absorbing water, I put a generally small up above it that just fit the size of the 3D printed pot.

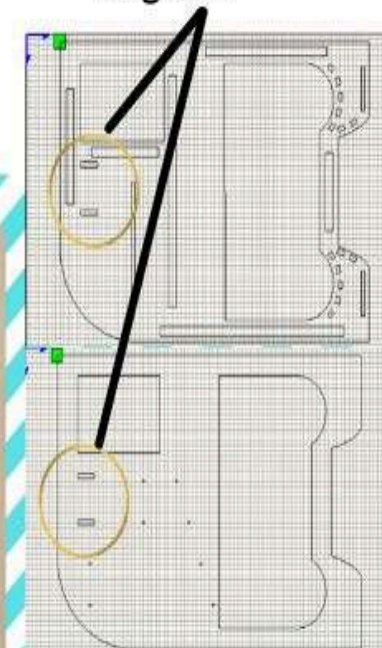


3D printing is another big project. Because I need to use Fusion 360 to model it, I have to learn which types of functions will avoid leaking, which means some techniques are not able to apply in Fusion.



I used various techniques to produce a project--a 3D box. It took 18 hours to print. after I send the code to the 3D printers, I set up the working place for it. The 3D printer creating an object with melted filaments. I need to put masking tapes on the working platform of the printer to make sure those plastic are easy to be removed later. After all, I put the black filament into the machine, start the machine and it begins its 18-hour printing process. It finished printing the next day. I removed the extra parts and I eventually get a smooth black plastic box with tenons at back.

It is assembled at the back of the shelf, which is a pot for the creeper to grow.



machine. When I can feel the plastic is cooled down and becomes hard again, I release it and the bend is formed. I do the same thing for the other sides of the box and finally, I get all the side bent. Then it is time for glueing. Making sure each edge are strong and perfect, I tape the edge first. Then I use silicon-based glue to glue two side together. after the glue is completely dry, I will get a perfect corner. However, the glue is not a transparent colour but white, which looks obvious.





Here I reach the third step--compare with the first two, it is quite simple. I identify the length of the metal strip with an addition bending length. I choose low carbon steel because of its flexibility. The sheet form of the LCS is incredibly soft, which I even can bend it with my hands. First, I asked technicians to use a circular saw to cut the entire sheet into separated strips with the same width and length. I choose 4 perfect strips and measure out the desired cutting and bending parts.

I use the masking tape to help me to mark out the specific lines and points for drilling and bending. The metal racks have different length, so I use the metal guillotine to cut off the unnecessary parts. I need to place the cutting edge right under the knife, which increases accuracy. The guillotine has a heavy top and a light bottom, so I need to step on its body to make it stay in the same position. Then I hold the strip, push the knife down slowly.

After cutting, it is necessary to smooth the sharp edges. Files and abrasive papers can make the edges blunt. I put each strip on vice and use the flat-file to smooth these sharp edges and corners.

It is time to drill holes for later fastening. I use a centre punch to create a dimple to guide the pillar drill, which can help the drill bit fix on that accurate position while drilling. Then, I clamp the strip carefully on the pillar drill's table with wood underneath. Keeping my hands away from the drill, wearing safety goggles to avoid other materials come into the eyes, these safety protections are essential throughout the entire process.

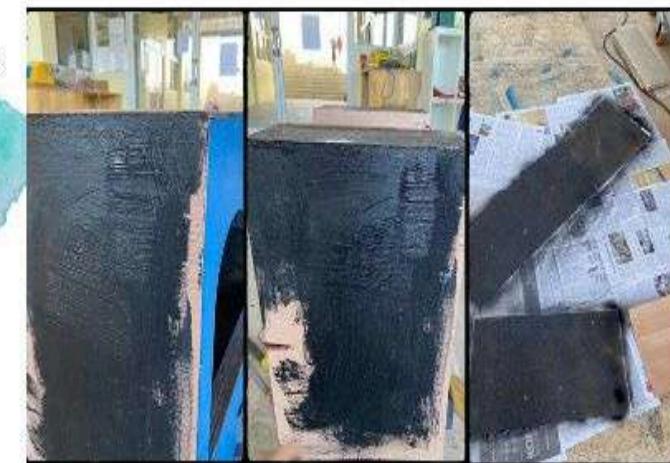
I remove the masking tapes on the drilling parts, leaving the bending lines only. I put the strip on the vice, using a ball-peen hammer to hit the metal powerfully, checking the angle of the bend side each time to make sure it is perfectly 90 degrees.



When all the strips are perfectly bent, they are ready for the final assembling.

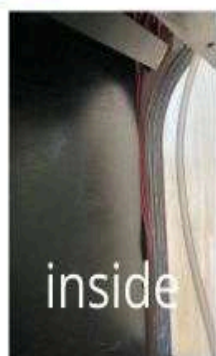


I also cut the flexible plywood while waiting for them to dry. The black spray paint cannot perform very well on the surface of the flexible plywood because of its rough grains. Thus, I choose acrylic paint. It can go through the grain of the wood, covering all the fissures on it. However, for the bottom and the curved part of the flexible plywood, I use spray paint because it is too large to paint with acrylic and it is not the obvious part of the whole project. I paint both inside and outside of the shelf, including the edges where the wooden pieces are glued together.



This is the last step before the final step. All the wooden components are needed to be spray painted to different colours. It is a way to make the project get its own 'skin colour'. Although the spray paint has a strong smell, which contains a little bit of toxic material, it is still the quick and easy option. It also has a variety of colour choices. I choose light blue and black spray paint. After the wood and metal are cut, including the flexible plywood later I will mention, I bring all of them downstairs to an opening area with good ventilation. Because the paint will release a strong smell with VOC(Volatile Organic Compounds), I must wear masks and aprons whilst painting. I put some old newspaper under my components in case the paint will colour the floor. Then, I stand back, keep the bottle away from the components about 1 meter and begin the painting process. Sometimes the components cannot be painted properly, so I need to spray paint them again. However, I try to decrease the use of paint since they are not eco-friendly.

After spray paint all of the wooden pieces, I leave them to dry for over one hour. During that time, I begin to deal with the metal strips. Though the metal is flexible, it doesn't resist rusting. I need to coat them with transparent paint. So I choose transparently metal primer to prevent these strips from rusting. I hang them in the air with a wire. I wear a mask while I spray the metal. It has some smell too so I open all the windows in the DT classroom in order to do it. The metal shines after painting, with a great smooth surface reflecting lights from all the direction. I am so pleased about that.





Assembling



After putting all the components together, the shelf looks already half finish. It is time to put the metal strips and the 3D printing creeper pot. I use the curve bending shape to help me to make sure all the metal curves are bent at the same angle and curve.

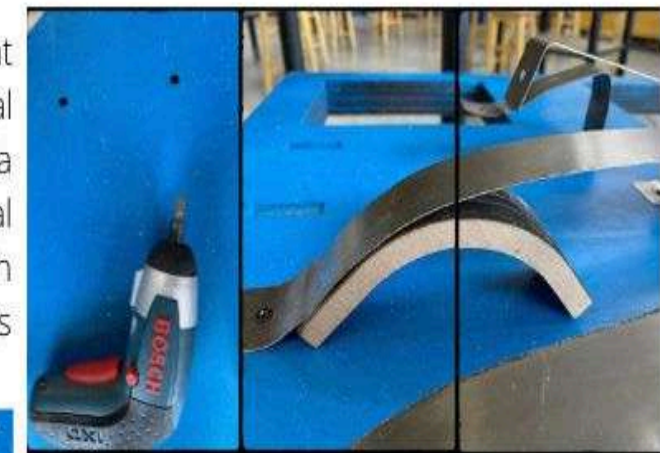
Finally, I will get on to assemble these wooden pieces. They are the overall structure of the whole project, with many mortises and tenons all around. Because they are measured and cut in the laser cutter, which gives it a super-accurate size and position. When I put them into the major front board, they are perfectly embedded together. However, when I up to the smaller pieces, they need some power to put inside just because their mortises have the exact size of the tenons. It can be either a good thing or a bad thing; the negativity is I spend a long time to push them in place; the positivity is they are staying in a sturdy position with each other.

After putting all the bigger pieces inside the front board, I start to put those small pieces for curves. For the curves outside of the board, I decided to use flexible plywood to wrap around. For the inside of the curves, I paint those small sticks black, using them on the top of the shelf for putting wires from lamps and solar panels later on.

Because the grow light is assembled on the top part of the major components, I screw it on first with four small flathead screws. It also has a long wire with a remote that can control the colour of the light and the time that the light will be on.

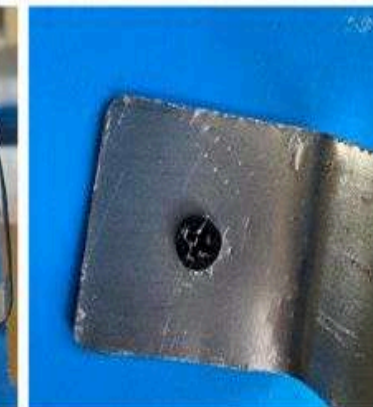


Different length of metal strips has a different position. It is hard to bend the short metal strip because the wood curve cannot hit with a mallet when it is squeezed under the metal strip. So I manage to push the wood piece with my hand and it is work, although the curve is not obvious.



With the finish of the metal strip's bending and assembling, it is time to put the 3D printed pot inside. I try to put it inside manually, but the size of the tenon on the pot is as same as the mortise on the board. So I need to use a filer to smooth the side of the mortise to make it bigger. It takes quite a while to make the mortises a little bigger than the tenons. I try to put the pot inside again with some strength and eventually it is embedded inside. Then I put the upcycling pot on it with the water-absorbing rope. It is the wick system for creeper. The black plastic pot is super fit for the shelf.

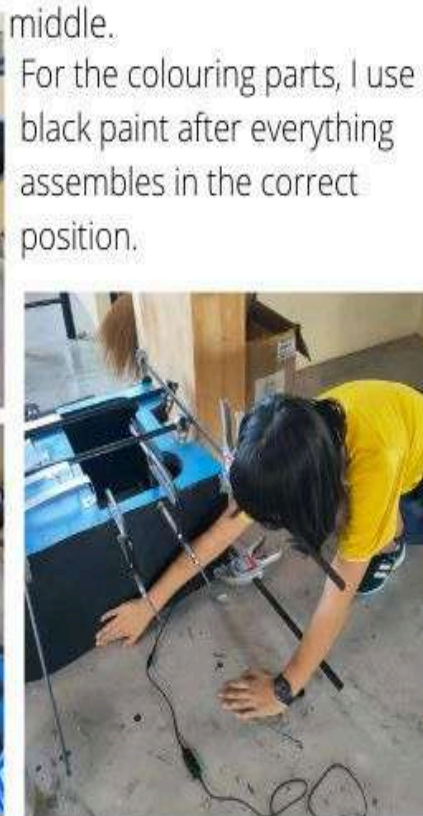
Luckily, I choose to buy these types of grow light before my project even get started, which provided me a long time to consider the way of dealing with wires. So I give the left component of the shelf a square gasp, letting the wires through there.



FLEXIBLE PLYWOOD



Moving on, it is time to assemble the flexible plywood! Because it is as thin as the veneer, I decide to glue some of them on the surface of the shelf to block the dark line from the laser cutter. I begin with the top curve first, using the PVA glue with box nails together. It is a great technique to make the flexible plywood stick on the shelf firmly. I also put a piece of plywood on the top board, making it has the same level of curve. After the top curve is stuck on the shelf, I am going to put another flexible wood on the bottom curve. It is a long piece, which requires a large amount of glue and clamps to make it stuck on the main components. The box nails are used around the edge of this large board and they need to be assembled on both edges to make sure the force is balanced on both ends. For the clamping process, I put other materials under the clamps to spread out the pressure.



PLANT POTS



The second thing I need to consider is the function of the flower pots. Deepwater culture needs a medium for plants to grow. At the same time, I prepare seeds at home for later use. I cut off sponges and make two things that I have to consider: The hole for the top as a cross for putting seeds inside. I use lettuce seeds. After a few days, because of the good temperature and humidity, most of the seeds have noticed, there is some extra gap on some of the board, which I cut them out in purpose to put wires through them. I measure the holes' position to make sure they are perfectly fit in the position. Then I cut them out with a band saw for the ones that have the holes on the side and with the chisel for the holes in the middle.



For the colouring parts, I use black paint after everything assembles in the correct position.

The last important thing is the solar panel. It is a power generator for a gas pumper for deep water culture. It has a long wire connected to a box with a motor and the box connects a gas tube. I already create the hole for wires to go through. Because its wire is super long, I manage to put them through the gap between the top right corner of the shelf. There is an empty place created by wooden strips and the upper boards. I also cut out a piece of flexible wood for wires to go through. Because the gas tube needs to close to the reservoir, I find the motor box is convenient to put at the top of the shelf--both wire and the gas tube are able to put in the correct position without affecting the appearance. For the solar panel, I put it one with flathead screws and washers. (the main reason is the panel have got a bigger hole than the screws)

SOLAR PANEL





THE SOLAR PANEL CAN WORK VERY WELL. I TEST IT IN THE FISH POT, WHICH IS EASIER TO SEE.

90 DEGREE HINGES CAN CHANGE THE DIRECTION OF THE SOLAR PANEL ACCORDING TO THE DIRECTION OF THE SUNLIGHT



AMAZING PLACES FOR CREEPERS TO GROW



PLACES TO PUT UTENSILS AND FISH POD



A HOLE TO WATCH THE WATER LEVEL

IT'S TIME TO EAT



evaluation

With the great care and supports from other teachers, the seeds finally grow up. The whole system still works very well. My teachers and I already try these freshly grown vegetables. Of course, they are super healthy and delicious. It is a successful product I would say. However, nothing is flawless. This shelf still has its own weaknesses.

Good water resistance for the reservoir. Upcycling is worked and fewer materials that are difficult to recycle are used(e.g. spray paint PVA glue)

It doesn't have special sharp edges and the reservoir is safe enough on the shelf, which will not cause problems. There are some specially designed curves as well.

This shelf has the same height as a 6-7 years old kid, which is convenient for people to reach if it is on a low coffee table or even on the floor.

This shelf includes. many different functions, which is amazing. The cost should between 1500-2500THB

It has plenty of functions, including main functions and all the optional functions. It has places for fish pots, which can be the nutrition for plants. The upcycling pots can either be places for growing or for germinating seeds.

The shelf has a shape of a bird(like an owl), which is a unique shape and the colour is modern. Blue and black with green plants is a great combination, symbolising water and nature.

The basic structure is easy to join, with both permanent joints and temporary joints.

It is a shelf full of different functions, which involve basic techniques so everyone can use it as long as he or she knows some knowledge of it.

base on the design brief and design specification, I notice:



The metal is not the one that really resists rusting. It can be improved with real rusting-resistant metal.

The shelf is a little heavy, which is not easy to bring around. However, it is a shelf means to have its own property of sturdy.

It is generally bigger in size, so it can be placed on either table or the floor.

It is a shelf for middle-class people so it is not a good choice for people who are living in a poor area. It should be simpler than this.

It could have some handles or wheels for the shelf to transport.

The colour is not that light, but there will be lots of choices if it is sell to the mass market.

This manufacture way involves many steps, which is not that simple But it can be disassembled into different pieces, which can be put in a flat-pack.

Because of its shape and weight, people who live in a flat might find it hard to carry it all the way up and downstairs.

future

Based on the product's quality and people's opinions, I think it is a sustainable idea for future farmland as well as a way to save the global issues. The style of the shelf should be various, using as a alternative gam

consultation

AFTER SHOWING THE FINAL OUTCOME TO MY CLIENTS AGAIN, THEY ALL GIVE ME DIFFERENT REPLIES.

CLIENT1&2:

WE LOVE IT. THE SIZE OF THE SHELF IS SUITED FOR OUR FLAT AND INDEED, IT IS MULTIFUNCTIONAL.

CLIENT 3&4&5:

IT IS AMAZING. DON'T FORGET WE COME UP WITH THIS IDEA TOGETHER! MAYBE NEXT TIME IT CAN HAVE MORE OTHER FUNCTIONS.



HUGE REFERENCES

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